



Investigation of the dynamic transmission accuracy of an industrial robot joint RV reducer under variable situations

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Abstract

The dynamic transmission error of the joint RV (rotating vector) reducer is one of the important factors affecting the positioning accuracy of industrial robots. However, the current studies on the transmission accuracy characteristics of RV reducers are based on the equal speed driving condition, which neglects the impact of the transmission error generated by the joint RV reducer on the positioning accuracy when the robot is in the start-stop variable speed phase. This study carries out an in-depth analysis of the dynamic transmission accuracy characteristics of joint RV reducers of industrial robots under variable-speed working conditions. Two typical mathematical models of variable-speed drive laws are constructed, and a multibody dynamic model of the RV reducer that takes into account the geometric errors of typical components is established, aiming to more accurately simulate the operating state of the RV reducer under actual working conditions. To confirm the accuracy of the theoretical RV reducer contact dynamics model, the dynamic transmission error test of the RV reducer under equal-speed conditions is carried out. The results show that the dynamic model can better reflect the actual transmission characteristics of the RV reducer. Thereafter, based on the theoretical model, the effects of different driving laws on the dynamic transmission error of joint RV reducers are analyzed in depth. The results of the study show that the transmission error of the RV reducer under variable speed drive is significantly larger than that under equal speed drive. Moreover, angular acceleration and load inertia are the main factors affecting the transmission error of RV reducers for industrial robot joints.

Keywords Multibody dynamic modeling · Joint RV reducers · Dynamic transmission error · Variable condition drive

1 Introduction

Apart from its extensive usage in welding, handling, and painting applications, industrial robots have also grown in significance in high-precision domains like precision manufacturing, aerospace, and precision measurement. Geometric errors and non-geometric errors are the two main underlying causes of motion errors in industrial robots [1]. The positioning

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and motion accuracy of industrial robots are directly impacted by geometric errors, which include joint reducer transmission faults, body manufacturing flaws, and assembly errors. Temperature fluctuations, control accuracy, and the structural flexibility of the robotic arm are the primary complicated affecting elements of the non-geometric errors [2]. Enhancing the motion precision performance of industrial robots through optimization design of the joint reducer's transmission precision performance has become a major priority in both academics and industry in recent years.

To improve the positioning accuracy of industrial robots, many scholars have conducted in-depth research on error compensation strategies. Holden et al. [3] adjusted its kinematic parameters by putting an offline error compensation approach into place. However, the effectiveness of offline compensation is limited in practice by other dynamic characteristics affecting the robot, such as clearance, dynamic load fluctuations, and thermal expansion. As a result, Sergio Ferrarini et al. [4] presented a generalized methodology for evaluating infrared attitude and path accuracy, focusing mainly on instrumentation and test procedures. A novel online compensation method is proposed and validated after an experimental campaign performed on a robot under different operating conditions. Liu et al. [5] proposed an innovative calibration method for five-degree-of-freedom hybrid robots, which combines the back-propagation neural network (BPNN) and the D-H method, and realizes high-precision real-time compensation by constructing an error prediction model and adopting a pose error decomposition strategy. This method effectively solves the conflict between measurement efficiency, calibration accuracy, and real-time compensation. However, the kinematic parameter errors in the robot manufacturing and assembly process have a direct impact on the positioning accuracy of the manipulator. Therefore, controlling the kinematic parameter errors within a reasonable range has become an urgent problem in the field of robotics, which relies on better localization accuracy analysis and tolerance design of robot manipulators [6]. Conventional tolerance design techniques frequently rely on the expertise of mechanical design manuals and designers, which is imprecise and unscientific [7, 8]. Several academics started attempting to incorporate contemporary design techniques into robot tolerance design to solve this issue. Among them, Liou et al. [9] standardized the tolerance ranges of robot kinematic parameters by using Taguchi's method to the tolerance design of robot manipulators. Rout and Mittal [10], on the other hand, conducted a sensitivity analysis of the positioning accuracy of a 2-degree-of-freedom planar robotic arm, considering the effects of factors such as mass, payload and torque on the positioning accuracy. Rao and Bhatti [11] used Monte Carlo simulation to assess the positioning precision reliability after developing a probabilistic model that included component dimensional tolerances and kinematic pair clearance.

Compensation through motion control to improve the positioning accuracy of industrial robots has been the primary focus of the above research. The inaccuracies caused by the joint transmission components also significantly affect the robot's accuracy. However, due to the fluctuation and poor consistency of the dynamic transmission errors of the joint transmission components, it is difficult to eliminate the effects of the transmission errors of the transmission components by means of motion compensation to improve the positioning accuracy of industrial robots. Rotate vector (RV) reducer is a fundamental part of industrial robots, and its performance directly affects the robot's total performance [12]. As shown in Fig. 1, RV reducers are commonly found in industrial robots located in high-load joints including the lumbar, shoulder, and elbow joints. Scholars have studied the RV reducer in detail, covering a variety of topics like cycloidal tooth profile design [13–15], optimization schemes [16], stress calculations [17], efficiency prediction [18], and transmission error analysis [18–21], in order to meet the demands in the field of high-precision transmission. Still, there has not

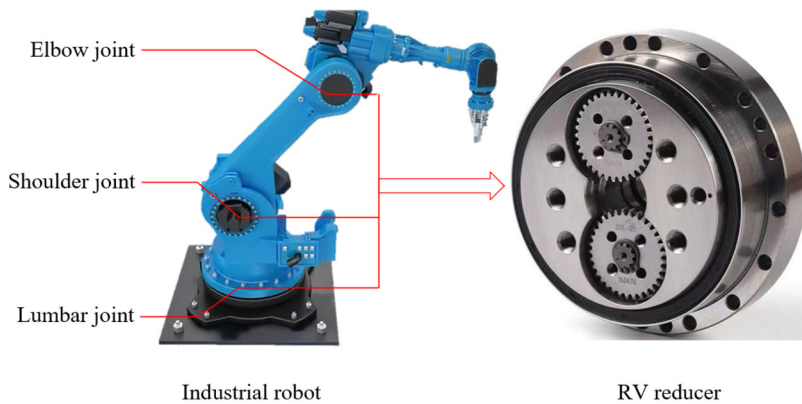


Fig. 1 RV reducers for an industrial robot

been enough research done on dynamic transmission error modeling and accuracy analysis of the RV reducer, even with all the attention paid to their overall performance. As a crucial metric indicating the precision of robot positioning, rotational transmission error modeling and analytic techniques require enhancement. Yang et al. [22] developed a correlation model between the original error and transmission accuracy, suggested an analytical error modeling approach for the RV reducer over-constrained structure, and offered theoretical justification for the accuracy enhancement of the RV reducer. Wang et al. [23] proposed a new multi-tooth contact and transmission error model for the RV reducer, which more accurately describes the transmission process of the reducer by dividing the contact area into multiple differential units. On the other hand, Ma et al. [24] used simulation to study the impact of many parameters on transmission error and compared the transmission accuracy with different RV reducers. An enhanced dynamic model of an RV reducer was suggested by Xu et al. [25] to investigate the moment stiffness and rotational accuracy of bearings planetary frame rotor system. Based on this, the moment stiffness and rotational accuracy of the bearings planetary frame rotor system were studied in relation to various load circumstances, axial pre-stress displacement, ball count, and combinations of the inner and outer ring groove radius of curvature factors.

Furthermore, the relationship between the reducer transmission error and various transmission component errors has not been thoroughly described in the aforementioned RV reducer error analysis model. As a result, a large number of academics have also conducted extensive research on the actual effects of various errors on the transmission accuracy of RV reducers. Ahn et al. [26] used the finite element approach to examine the tolerance and friction of an RV reducer to investigate how these factors affect the output torque of the reducer as well as its many contacts. More specifically, friction increases the output torque variation, whereas tolerance mostly lowers the torsional rigidity. Taking into account the unpredictability of manufacturing error and assembly mistake, Jiang et al. [27] examined the dynamic transmission performance of the RV reducer using a theoretical model, a virtual prototype simulation, and a physical prototype test, respectively. Wang et al. [28] proposed a new method to study the transmission accuracy of the RV reducer using the response surface method (RSM). On the basis of constructing a virtual prototype of the RV reducer, the influence of the error of individual transmission components on the overall transmission error of the reducer was analyzed.

Dynamic experiments are extremely important to thoroughly study the dynamic transmission accuracy performance of the RV reducer. Several researchers have tested the RV reducer in-depth and examined how various fault statuses, load levels, and speed affect the reducer's performance. Xu et al. [29] experimentally tested the transmission error of the RV reducer under different working conditions, revealing the influence law of load and speed on the transmission error. Furthermore, Xu et al. [30] cleverly integrated the precision reducer's Stribeck friction model and presented a novel approach for figuring out the ideal measurement speed. For the RV reducer, experiments with transmission error measuring at various speeds were conducted. The peak value of the RV reducer's transmission error measurement is minimized at the ideal measurement speed, hence minimizing the impact of rotational speed on the measurement outcomes. Conversely, Pietro Bilancia et al. [31] suggested new experimental configurations and techniques with the goal of enhancing the robot's accuracy performance. By designing control techniques to adjust for torque variations and input speed inaccuracies, they further refined the transmission accuracy performance of the RV reducer and sought to improve the accuracy of the evaluation of the transmission performance of dynamic servo-mechanisms.

Although the transmission accuracy analysis under isokinetic conditions is the primary focus of the above research on the industrial robot joint RV reducer, industrial robot joints frequently experience start-stop acceleration and deceleration movements in practical applications, and their output load varies in tandem with the movements of the robots. It is obvious that the joint RV reducer's simulated transmission error law under equal speed settings cannot accurately represent the transmission accuracy features of industrial robots operating under load and speed variations. The main contribution of this research is to study the dynamic transmission accuracy characteristics of the RV reducer for industrial robot joints under variable operating conditions. First, the industrial robot joint RV reducer's dynamic transmission error model and the mathematical model of its two typical driving laws are built, and the model's typical error is taken into account. Then, through theoretical simulation and testing, the model's accuracy is confirmed. Lastly, the impacts of velocity, angular acceleration, and load inertia on the joint reducer's accuracy of transmission are examined and investigated, based on various variable speed driving rules.

2 Joint drive law for industrial robots

This section builds two typical mathematical models of acceleration and deceleration driving laws to thoroughly examine the transmission error of the RV reducer of industrial robots operating under variable-speed conditions. They are the equal acceleration and equal deceleration (EAED) driving law, which is the trapezoidal acceleration and deceleration driving law [32], and the sinusoidal acceleration and deceleration (SAD) driving law, which is the 'S' curve acceleration and deceleration driving law [33].

2.1 Equal acceleration and equal deceleration driving law

In the EAED driving law, its acceleration and deceleration phase acceleration value is constant, and the speed increases or decreases linearly with the increase of time. Robot joints, driven by the law of EAED, include seven phases in a cycle: forward acceleration, forward uniform velocity, forward deceleration, work point dwell, reverse acceleration, reverse uniform velocity, and reverse deceleration, as shown in Fig. 2. The maximum allowable input

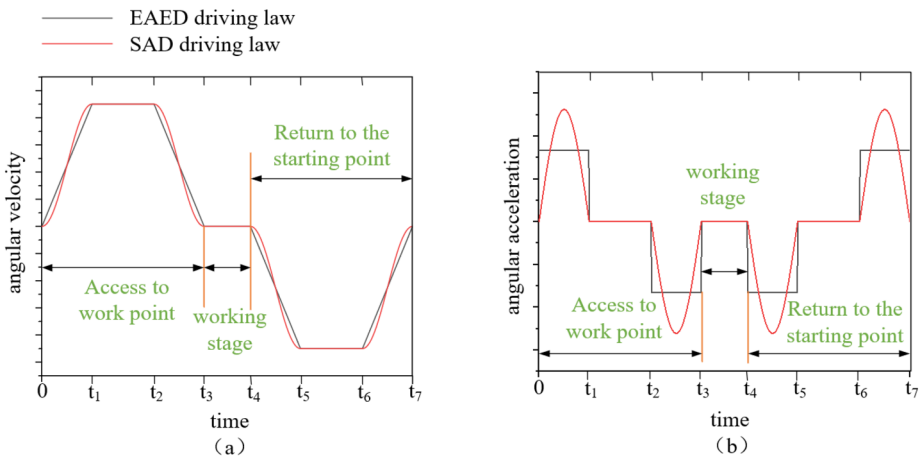


Fig. 2 Parameterized mathematical models of two typical acceleration and deceleration drive laws: (a) curves of angular velocity; (b) curves of angular acceleration

angular velocity $\omega_{\max}$, the joint output angle θ_0 , and the duration of a single work cycle t_s determine how much time is allotted to each of the seven phases.

Based on the joint reducer ratio, the input angle θ can be calculated as

$$\theta = \theta_0 \cdot i_{s-p}, \quad (1)$$

where i_{s-p} is the speed ratio of the RV reducer at the joints of the industrial robot.

The time of each phase of the EAED driving law is shown in Eq. (2):

$$\begin{cases} t_1 = (t_s - t_0)/2 - t_2 \\ t_2 = \theta / \omega_{\max} \\ t_3 = (t_s - t_0)/2 \\ t_4 = t_3 + t_0 \\ t_5 = t_4 + t_1 \\ t_6 = t_5 + t_2 - t_1 \\ t_7 = t_6 + t_1 \end{cases}, \quad (2)$$

where t_2 is calculated by combining Eq. (3) and Eq. (4); t_s is the total time of a cycle, and t_0 is the dwell time at the robot work point. Equation (3) is the formula for calculating the angular acceleration of the EAED driving law; Eq. (4) is the formula for calculating the angular displacement of the EAED driving law.

$$a_1 = \omega_{\max} / t_1. \quad (3)$$

$$2 \times \frac{1}{2} a_1 t_1^2 + \omega_{\max} (t_2 - t_1) = \theta. \quad (4)$$

As seen in Fig. 2(b), the acceleration expression corresponding to each stage is displayed in Eq. (5) and Eq. (3):

$$a(t) = \begin{cases} a_1 & 0 \leq t < t_1 \\ 0 & t_1 \leq t < t_2 \\ -a_1 & t_2 \leq t < t_3 \\ 0 & t_3 \leq t < t_4 \\ -a_1 & t_4 \leq t < t_5 \\ 0 & t_5 \leq t < t_6 \\ a_1 & t_6 \leq t < t_7 \end{cases} \quad (5)$$

Integration of Eq. (5) with respect to time t yields angular velocity as shown in Eq. (6):

$$\omega(t) = \begin{cases} a_1 t & 0 \leq t < t_1 \\ \omega_{\max} & t_1 \leq t < t_2 \\ \omega_{\max} - a_1(t - t_2) & t_2 \leq t < t_3 \\ 0 & t_3 \leq t < t_4 \\ -a_1(t - t_4) & t_4 \leq t < t_5 \\ -\omega_{\max} & t_5 \leq t < t_6 \\ -\omega_{\max} + a_1(t - t_6) & t_6 \leq t < t_7 \end{cases} \quad (6)$$

Integrating the angular velocity with respect to time t of Eq. (6) yields the angular displacement as shown in Eq. (7) and Eq. (8):

$$\theta(t) = \begin{cases} \frac{1}{2}a_1 t^2 & 0 \leq t < t_1 \\ \theta_1 + \omega_{\max}(t - t_1) & t_1 \leq t < t_2 \\ \theta_2 + \omega_{\max}(t - t_2) - \frac{1}{2}a_1(t - t_2)^2 & t_2 \leq t < t_3 \\ \theta_3 & t_3 \leq t < t_4 \\ \theta_4 - \frac{1}{2}a_1(t - t_4)^2 & t_4 \leq t < t_5 \\ \theta_5 - \omega_{\max}(t - t_5) & t_5 \leq t < t_6 \\ \theta_6 - \omega_{\max}(t - t_6) + \frac{1}{2}a_1(t - t_6)^2 & t_6 \leq t < t_7 \end{cases} \quad (7)$$

The input angles for each stage are as follows:

$$\begin{cases} \theta_1 = \frac{1}{2}a_1 t_1^2 \\ \theta_2 = \theta_1 + \omega_{\max}(t_2 - t_1) \\ \theta_3 = \theta_2 - \frac{1}{2}a_1(t_3 - t_2)^2 \\ \theta_4 = \theta_3 \\ \theta_5 = \theta_4 - \frac{1}{2}a_1(t_5 - t_4)^2 \\ \theta_6 = \theta_5 - \omega_{\max}(t_6 - t_5) \\ \theta_7 = \theta_6 - \omega_{\max}(t_7 - t_6) + \frac{1}{2}a_1(t_7 - t_6)^2 \end{cases} \quad (8)$$

Due to the fixed acceleration and deceleration stage acceleration value of the EAED drive law, the robot requires the least amount of time to complete a work cycle when operating under the same motion parameters but with a different variable speed drive law. This makes it appropriate for applications requiring high efficiency from robots. However, from the angular acceleration change curve of the EAED driving law it can be seen that there is a step phenomenon in its angular acceleration value at the interface between the variable speed stage and the uniform speed stage, and at the starting and stopping stages, as shown in Fig. 2(b). This will cause a great impact on the robot joints and reduce the smoothness of robot operation.

2.2 Sinusoidal acceleration and deceleration driving law

In the SAD driving law, its acceleration value is continuously and smoothly changing and there is no step phenomenon. Its velocity increases or decreases nonlinearly with time. Robot joints driven by SAD also consist of seven phases in a cycle: forward acceleration, forward uniform velocity, forward deceleration, work point dwell, reverse acceleration, reverse uniform velocity, and reverse deceleration, as shown in Fig. 2. The time allocation of the seven phases is self-allocated according to the joint output angle θ_0 , the maximum allowable input angular velocity $\omega_{\max}$, and the total time of one cycle of work t_s .

The input angle θ is determined by the output angle and the reducer ratio, as in Eq. (1). The time allocation for each phase of the SAD driving law is shown in Eq. (2). As seen in Fig. 2(b), the acceleration expression corresponding to each stage is displayed in Eq. (9):

$$a(t) = \begin{cases} \frac{\pi\theta}{2t_1t_2} \cdot \sin\left(\frac{\pi}{t_1} \cdot t\right) & 0 \leq t < t_1 \\ 0 & t_1 \leq t < t_2 \\ \frac{\pi\theta}{2t_1t_2} \cdot \sin\left(\frac{\pi}{t_1} \cdot t - (t_2 - t_1) \cdot \frac{\pi}{t_1}\right) & t_2 \leq t < t_3 \\ 0 & t_3 \leq t < t_4 \\ \frac{\pi\theta}{2t_1t_2} \cdot \sin\left(\frac{\pi}{t_1} \cdot t - (t_4 - t_1) \cdot \frac{\pi}{t_1}\right) & t_4 \leq t < t_5 \\ 0 & t_5 \leq t < t_6 \\ \frac{\pi\theta}{2t_1t_2} \cdot \sin\left(\frac{\pi}{t_1} \cdot t - t_6 \cdot \frac{\pi}{t_1}\right) & t_6 \leq t < t_7 \end{cases} \quad (9)$$

Integration of Eq. (9) with respect to time t yields angular velocity as shown in Eq. (10):

$$\omega(t) = \begin{cases} -\frac{\theta}{2t_2} \cdot \cos\left(\frac{\pi}{t_1} \cdot t\right) + \frac{\theta}{2t_2} & 0 \leq t < t_1 \\ \omega_{\max} & t_1 \leq t < t_2 \\ \omega_{\max} + \left(-\frac{\theta}{2t_2} \cdot \cos\left(\frac{\pi}{t_1} \cdot t - (t_2 - t_1) \cdot \frac{\pi}{t_1}\right) - \frac{\theta}{2t_2}\right) & t_2 \leq t < t_3 \\ 0 & t_3 \leq t < t_4 \\ -\frac{\theta}{2t_2} \cdot \cos\left(\frac{\pi}{t_1} \cdot t - (t_4 - t_1) \cdot \frac{\pi}{t_1}\right) - \frac{\theta}{2t_2} & t_4 \leq t < t_5 \\ -\omega_{\max} & t_5 \leq t < t_6 \\ -\omega_{\max} + \left(-\frac{\theta}{2t_2} \cdot \cos\left(\frac{\pi}{t_1} \cdot t - \frac{\pi}{t_1} \cdot t_6\right) + \frac{\theta}{2t_2}\right) & t_6 \leq t < t_7 \end{cases} \quad (10)$$

Integrating the angular velocity with respect to time t of Eq. (10) yields the angular displacement as shown in Eq. (11) and Eq. (12):

$$\theta(t) = \begin{cases} -\frac{t_1\theta}{2\pi t_2} \cdot \sin(\frac{\pi}{t_1} \cdot t) + \frac{\theta}{2t_2} & t_0 \leq t < t_1 \\ \theta_1 + \omega_{\max}(t - t_1) & t_1 \leq t < t_2 \\ \theta_2 + \omega_{\max}(t - t_2) & t_2 \leq t < t_3 \\ +(-\frac{t_1\theta}{2t_2\pi} \cdot \sin(\frac{\pi}{t_1} \cdot t - (t_2 - t_1) \cdot \frac{\pi}{t_1}) - \frac{\theta}{2t_2} \cdot (t - t_2)) & t_3 \leq t < t_4 \\ \theta_3 & t_4 \leq t < t_5 \\ \theta_4 - \frac{t_1\theta}{2t_2\pi} \cdot \sin(\frac{\pi}{t_1} \cdot t - (t_4 - t_1) \cdot \frac{\pi}{t_1}) - \frac{\theta}{2t_2} \cdot (t - t_4) & t_5 \leq t < t_6 \\ \theta_5 - \omega_{\max}(t - t_5) & t_6 \leq t < t_7 \\ \theta_6 - \omega_{\max}(t - t_6) + (-\frac{t_1\theta}{2t_2\pi} \cdot \sin(\frac{\pi}{t_1} \cdot t - t_6 \cdot \frac{\pi}{t_1}) + \frac{\theta}{2t_2} \cdot (t - t_6)) & \end{cases} \quad (11)$$

The input angles for each stage are as follows:

$$\begin{cases} \theta_1 = \frac{\theta}{2t_2} \cdot t_1 \\ \theta_2 = \theta_1 + \omega_{\max}(t_2 - t_1) \\ \theta_3 = \theta_2 - (-\frac{t_1\theta}{2t_2\pi} \cdot \sin(\frac{\pi}{t_1} \cdot t_3 - (t_2 - t_1) \cdot \frac{\pi}{t_1}) - \frac{\theta}{2t_2} \cdot (t_3 - t_2)) \\ \theta_4 = \theta_3 \\ \theta_5 = \theta_4 - \frac{t_1\theta}{2t_2\pi} \cdot \sin(\frac{\pi}{t_1} \cdot t_5 - (t_4 - t_1) \cdot \frac{\pi}{t_1}) - \frac{\theta}{2t_2} \cdot (t_5 - t_4) \\ \theta_6 = \theta_5 - \omega_{\max}(t_6 - t_5) \\ \theta_7 = \theta_6 - (-\frac{t_1\theta}{2t_2\pi} \cdot \sin(\frac{\pi}{t_1} \cdot t_7 - t_6 \cdot \frac{\pi}{t_1}) + \frac{\theta}{2t_2} \cdot (t_7 - t_6)) \end{cases} \quad (12)$$

Compared with the EAED driving law, the angular acceleration of the SAD driving law does not have the step phenomenon, so it can reduce the effects of the impact and improve the smoothness of the robot's working process, so the SAD driving law is generally used in the occasion of the higher speed. However, compared to the same motion parameters and alternative variable-speed drive laws, its time consumed is quite considerable, which lowers the robot efficiency since its angular acceleration value fluctuates from small to large or large to small during the variable-speed phase.

3 Dynamic transmission error modeling of RV reducers for industrial robot joint

3.1 Transmission principle and structure of the industrial robot joint RV reducer

In modeling, the RV-40E reducer (the reducer is from Nabtesco Corp) is used as the object of study. Figure 3(a) shows the transmission principle of the RV reducer, while Fig. 3(b) provides its transmission structure. There are three types of bearings included in the RV-40E, which are swivel arm bearings (needle roller bearings), support bearings (tapered roller bearings), and large bearings (angular contact ball bearings). The swivel arm bearing is located in the eccentric of the crankshaft. This bearing does not have separate inner and outer ring, the crankshaft eccentric is the inner raceway of the swivel arm bearing, and

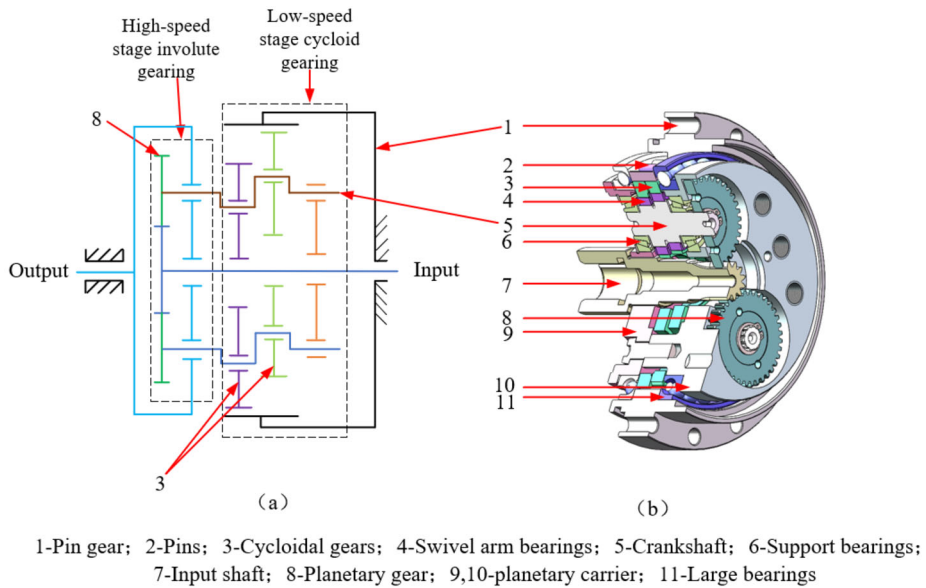


Fig. 3 Industrial robot joint RV reducer: (a) Transmission principle; (b) Structural design

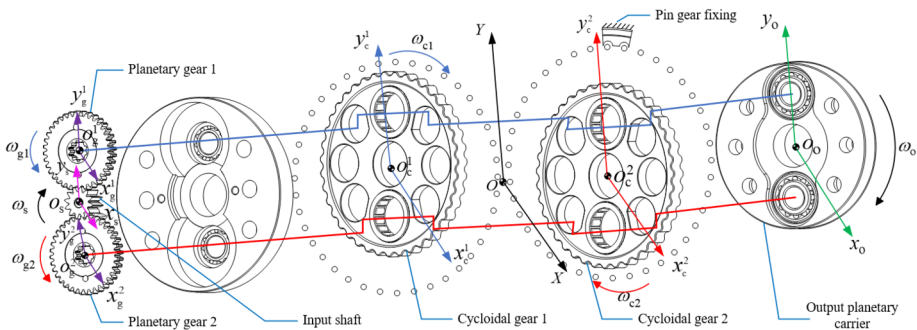


Fig. 4 The main kinematic rigid-bodies in the dynamic model of the RV-40E reducer

the cycloidal gear bearing holes are the outer raceway of the swivel arm bearing, which creates relative rotation between the crankshaft and the two pieces of the cycloidal gears. The support bearing is located at the crankshaft end step location and forms the relative rotation between the crankshaft and the two planetary carriers. The large bearing is located between the planetary carrier and the pin gear, forming a relative rotation between the planetary carrier and the pin gear.

The RV reducer is a two-stage transmission, its principle of transmission has been described in detail in many literatures and can be found in reference [22].

3.2 Multibody dynamics modeling of the RV reducer for industrial robot joint

Based on the multibody dynamics theory, a general approach for dynamic modeling of the RV-40E reducer was presented, as illustrated in Fig. 4. The dynamic model takes into

account the structure of high-speed stage involute gear transmission and low-speed stage cycloidal-pin transmission, and considers the influence of the support bearings at both ends of the crankshaft and the swivel arm bearings at the eccentric circles. In this dynamics model, a crankshaft with a planetary gear located on it is considered as one independent kinematic component. A total of six kinematic entities are included, respectively an input sun gear, two planetary gears and crankshaft combinations, two pieces of cycloidal gears, and an integral planetary carrier. As shown in Fig. 4, the input sun gear's angular velocity is ω_s , the planetary gear's angular velocity is ω_g , the two pieces of cycloidal gears' angular velocities are ω_{c1} and ω_{c2} , and the output planetary carrier's angular velocity is ω_o .

The transmission system of the dynamic model includes 18 generalized coordinates, represented as follows:

$$\mathbf{q} = (x_s, y_s, \theta_s, x_g^1, y_g^1, \theta_g^1, x_g^2, y_g^2, \theta_g^2, x_c^1, y_c^1, \theta_c^1, x_c^2, y_c^2, \theta_c^2, x_o, y_o, \theta_o), \quad (13)$$

where (x_s, y_s, θ_s) is the input sun gear's generalized coordinate; $(x_g^1, y_g^1, \theta_g^1)$ and $(x_g^2, y_g^2, \theta_g^2)$ are the two planetary gear and crankshaft assemblies' generalized coordinates; $(x_c^1, y_c^1, \theta_c^1)$ and $(x_c^2, y_c^2, \theta_c^2)$ are the two cycloidal gears' generalized coordinates; (x_o, y_o, θ_o) is the planetary carrier assemblies' generalized coordinate.

The transmission system mass matrix is represented as follows:

$$\mathbf{M} = \text{diag}(m_s, m_s, I_s, m_g, m_g, I_g, m_g, m_g, I_g, m_c, m_c, I_c, m_c, m_c, I_c, m_o, m_o, I_o), \quad (14)$$

where mass and moment of inertia of the input sun gear are indicated by m_s and I_s , respectively; mass and moment of inertia of the planetary gear and crankshaft assembly are indicated by m_g and I_g , respectively; mass and moment of inertia of the cycloidal gears are indicated by m_c and I_c , respectively; mass and moment of inertia of the planetary carrier assembly are indicated by m_o and I_o , respectively.

The driveline constraint equations for the isokinetic condition are expressed as follows:

$$\Phi = \begin{Bmatrix} x_s \\ y_s \\ x_o \\ y_o \\ \theta_s - \omega_s t \end{Bmatrix} = 0. \quad (15)$$

The driveline constraint equations for variable velocity conditions are expressed as follows:

$$\Phi = \begin{Bmatrix} x_s \\ y_s \\ x_o \\ y_o \\ \theta_s - \int_{t_i}^{t_{i+1}} \int_{t_i}^{t_{i+1}} a_i dx dt \quad t_i < t \leq t_{i+1} \end{Bmatrix} = 0, \quad (16)$$

where a_i is the angular accelerations of the variable speed phases; t_i is the time points of the phases of the variable speed process. The purpose of double integration of the angular acceleration with respect to time t is to obtain the angular displacement at that time period.

With the support of the above constraint equations, the input sun gear and the planetary carrier of the RV reducer are limited to rotational degrees of freedom, and the input sun gear requires that it revolves at a phase-appropriate angular velocity. The dynamic contact

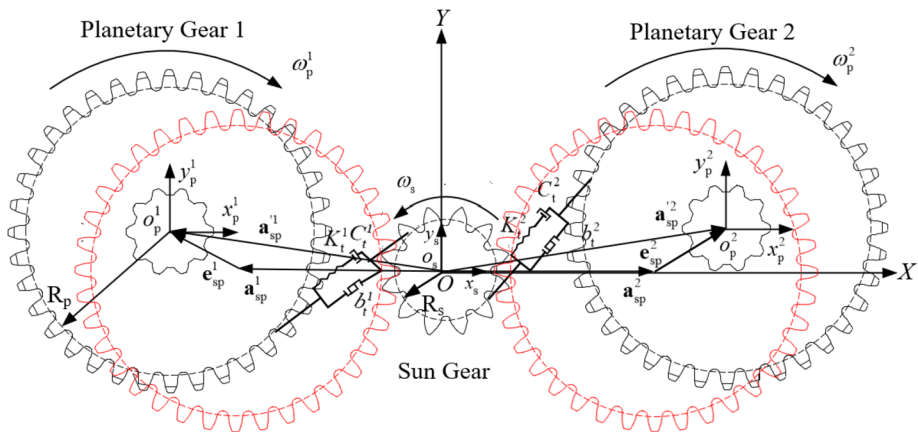


Fig. 5 Contact dynamics modeling of high-speed stage planetary gearing

traits between the components in the transmission system dictate the kinematics of other components.

Utilizing Baumgarte's approach [34], the dynamic equations of the transmission system are expressed as follows:

$$\begin{bmatrix} \mathbf{M} & \Phi_q^T \\ \Phi_q & \mathbf{0} \end{bmatrix} \begin{bmatrix} \ddot{\mathbf{q}} \\ \lambda \end{bmatrix} = \begin{bmatrix} \mathbf{Q}^G \\ \gamma - 2\alpha\dot{\Phi} - \beta^2\Phi \end{bmatrix}, \quad (17)$$

where $\gamma = -(\Phi_q\dot{\mathbf{q}})_q\dot{\mathbf{q}} - 2\Phi_{qt}\dot{\mathbf{q}} - \Phi_{tt}$; Φ_q is the Jacobi matrix of constraint equations; $\dot{\mathbf{q}}$ is the generalized velocity vector of the system; $\ddot{\mathbf{q}}$ is the generalized acceleration vector of the system; λ is the Lagrange multiplier; $\mathbf{Q}^G$ is the generalized force vector of the system, which generally includes gravity, and the contact force between contact components; Φ_{qt} is the Jacobi matrix of constraint equations with respect to the time derivation; Φ_{tt} is the second-order derivation of the constraint equations with respect to the time. The terms $2\alpha\dot{\Phi}$ and $\beta^2\Phi$ in (17) play the role of a control term.

3.3 Contact modeling of high-speed stage involute gear transmission

The contact dynamics model of the planetary gears of the RV-40E reducer is shown in Fig. 5, and equal springs are used to imitate the meshing interaction between the gears. The coordinate OXY in the figure is the global coordinate, $o_sx_sy_s$ is the local coordinate of the input sun gear, $o_g^1x_g^1y_g^1$ and $o_g^2x_g^2y_g^2$ are the local coordinates of the planetary gear and crankshaft assemblies. The gear mesh stiffness is denoted by K_t , the gear mesh damping is denoted by C_t , and the dynamic backlash on the tooth side of the gear pair in the transmission is denoted by b_t . The input sun gear reference circle radius is denoted by R_s , the planetary gear reference circle radius is denoted by R_g . The time-varying characteristics of the actual center distance of the gear transmission result from the elastic runout in the planetary gear's center due to the consideration of the support bearing's elastic influence on the crankshaft. Therefore, $\mathbf{a}_{sg}$ in the figure represents the ideal center distance vector between the sun gear and the planetary gear, $\mathbf{a}'_{sg}$ represents the actual center distance vector between the sun gear and the planetary gear, and $\mathbf{e}_{sg}$ represents the radial runout position vector of the planetary gear center. In the RV-40E reducer model, two sets of planetary gears are included, so the above parameters also contain two sets, differentiated by 1 and 2, as shown in Fig. 5 [35].

Since the meshing phases and the planetary gear center runout are not exactly the same in two pairs of gears, the values of the tooth flank dynamic backlash b_t in the model will be different. According to the nature of involute geometry, the dynamic backlash of the gear pair b_t is expressed as follows:

$$b_t = \left(\frac{2\pi R'_s}{Z_s} - s'_s - s'_g \right) \cos \alpha' + b_o, \quad (18)$$

where R'_s is the radius of the pitch circle when the sun gear meshes with the planetary gear; Z_s is the tooth count of the sun gear; s'_s and s'_g are the thicknesses of the pitch circle teeth of the sun gear and the planetary gear, respectively; α' is the meshing angle; b_o is the tooth's initial backlash.

The tooth thickness of the sun gear and planetary gear can be obtained from the equation of tooth thickness of any circle as follows:

$$s'_{s,g} = s_{s,g} \frac{R'_{s,g}}{R_{s,g}} - 2R'_{s,g} (\text{inv}(\alpha') - \text{inv}(\alpha)), \quad (19)$$

where $R'_{s,g}$ is the radius of the pitch circle of the sun and planet gears; $R_{s,g}$ is the radius of the reference circle of the sun and planet gears; $s_{s,g}$ is the tooth thickness of the reference circle of the sun and planet gears; and α is the pressure angle of the reference circle; inv is the involute function, $\text{inv}(\alpha_k) = \tan(\alpha_k) - \alpha_k$, α_k represents the pressure angle at any point on the involute.

The tooth thickness of the reference circle of the sun and planet gears is calculated as follows:

$$s_s = s_g = \frac{m\pi}{2}, \quad (20)$$

where m is the gear module.

The results calculated from the relationship between the pitch circle radius and the reference circle radius are as follows:

$$R_{s,g} \cos \alpha = R'_{s,g} \cos \alpha'. \quad (21)$$

The radius of the reference circle of the sun and planet gears is calculated as follows:

$$R_{s,g} = \frac{mZ_{s,g}}{2}. \quad (22)$$

Substituting Eq. (20), (21), and (22) into Eq. (19) yields

$$s'_{s,g} = \frac{\pi R'_s}{Z_s} - 2R'_{s,g} (\text{inv}(\alpha') - \text{inv}(\alpha)). \quad (23)$$

Calculated from non-standard center distance as follows:

$$a' = R'_s + R'_g. \quad (24)$$

Substituting Eq. (23) and Eq. (24) into Eq. (18) yields

$$b_t = 2a' \cos(\alpha') (\text{inv}(\alpha') - \text{inv}(\alpha)) + b_o. \quad (25)$$

The angle of meshing and the actual center distance have the following relationship:

$$\alpha' = \arccos\left(\frac{a}{a'} \cos \alpha\right), \quad (26)$$

where a is the ideal center distance of the gear;

The gear pair's dynamic tooth backlash can be stated as follows:

$$b_t = 2a \cos(\alpha)(\text{inv}(\alpha') - \text{inv}(\alpha)) + b_0. \quad (27)$$

The following expression can be used to express the relative displacement in the direction of the gear mesh line:

$$g_t = (R'_s \theta_s - R'_g \theta_g) \cos \alpha' + e_x \cos \alpha' + e_y \sin \alpha', \quad (28)$$

where R'_g is the radius of the pitch circle when the planetary gear meshes with the sun gear; e_x and e_y represent the runout of the planetary gear center in the direction of the coordinate axis.

The tooth side clearance function f_t , which can be stated as follows, is used to ascertain the gear pair's contact status during meshing:

$$f_t = \begin{cases} g_t & g_t \geq 0 \\ 0 & -b_t < g_t < 0 \\ |g_t + b_t| & g_t \leq -b_t \end{cases} \quad (29)$$

When $g_t \geq 0$, the sun gear and the planetary gear are in tooth contact force transmission; when $-b_t < g_t < 0$, the gears are in disengaged contact non-force transmission; when $g_t \leq -b_t$, the gears are in tooth back contact force transmission.

The expression method of gear mesh stiffness in acceleration and deceleration stages is relatively complex due to the time-varying of the meshing frequency in the case of variable speed. To simplify the calculation, the gear meshing stiffness K_t is set as a constant in the model, and its value is replaced by the average value of the time-varying gear meshing stiffness corresponding to the average rotational speed during the speed change process. The detailed calculation of the time-varying meshing stiffness of gears can be found in literature [36].

Gear pair mesh damping C_t varies with tooth backlash and is expressed as follows:

$$C_t = \begin{cases} 0 & f_t < d_0 \\ C_m \frac{f_t - d_0}{d_1 - d_0} & d_0 \leq f_t \leq d_1 \\ C_m & f_t > d_1 \end{cases}, \quad (30)$$

where C_m is a given value of meshing damping; d_0 and d_1 are critical values of contact depth, $d_1 > d_0$.

The gear pair's dynamic contact meshing force can be expressed as follows:

$$F = K_t (f_t)^{10/9} + C_t \dot{g}_t, \quad (31)$$

where the meshing line's relative velocity is represented by $\dot{g}_t$.

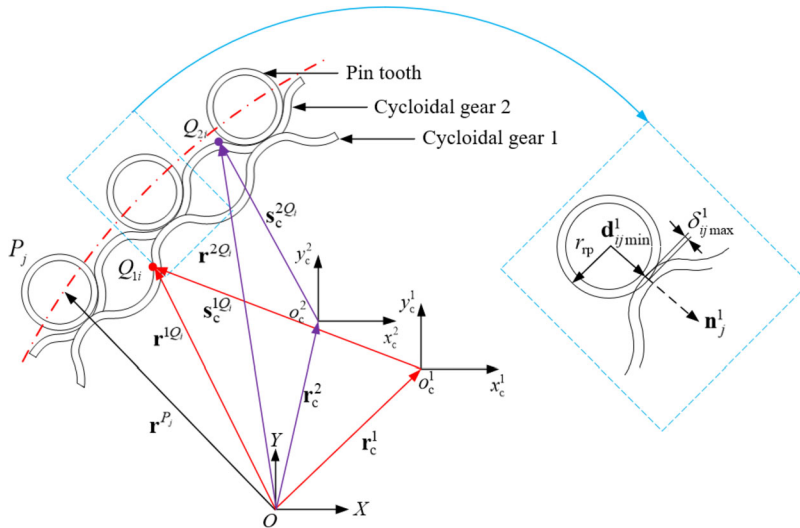


Fig. 6 Modeling the contacts of the low-speed stage cycloidal-pin transmission

3.4 Contact modeling of low-speed stage cycloidal-pin transmission

The contact dynamics model of the low-speed stage of the RV-40E reducer is shown in Fig. 6. In the figure, coordinate $o_c x_c y_c$ is the local coordinate of the cycloidal gears, which is located at the geometric center of the cycloidal gears. In the generalized coordinate, the position vector of each pin tooth center P_j is $\mathbf{r}^{P_j}$. The position vector of point Q_i on the tooth profile of the cycloidal gears is represented as follows:

$$\mathbf{r}^{Q_i} = \mathbf{r}_c + \mathbf{A}_c \mathbf{s}_c^{Q_i}, \quad (32)$$

where $\mathbf{r}_c$ is the origin position vector of the cycloidal gears' local coordinate in the generalized coordinate; $\mathbf{s}_c^{Q_i}$ is the contour point position vector of the cycloidal teeth in the localized coordinate; $\mathbf{A}_c$ is the coordinate transformation matrix for describing the orientation of the local coordinate of the cycloidal gears in the generalized coordinate, as shown in Eq. (33); i is the serial number of the cycloidal tooth contour discrete point.

$$\mathbf{A}_c = \begin{bmatrix} \cos(\theta_c) & -\sin(\theta_c) \\ \sin(\theta_c) & \cos(\theta_c) \end{bmatrix}, \quad (33)$$

where θ_c is the cycloidal gear's local coordinate's rotational angle in relation to its generalized coordinate.

The following expression represents the relative position vectors between the center of each pin tooth and the tooth profile of the cycloidal gears:

$$\mathbf{d}_{ij} = \mathbf{r}^{Q_i} - \mathbf{r}^{P_j}, \quad (34)$$

where j is the pin tooth serial number.

In the dynamic calculation, Eq. (34) is executed $i \times j$ times to cycle the calculation, and the contact analysis between the cycloidal gear and the pin tooth under a certain instant of time can be completed.

The conditions for determining whether or not contact is formed between the cycloidal tooth profile and each pin tooth are as follows:

$$\|\mathbf{d}_{ij}\| - r_{tp} \leq 0, \quad (35)$$

where r_{tp} is the pin tooth radius.

After contact is formed, the elastic contact deformation can be calculated using the following equation:

$$\delta_{ij} = \|\mathbf{d}_{ij}\| - r_{tp}. \quad (36)$$

During the contact analysis process, which screens for the maximum contact deformation $\delta_{ij \max}$ between each pin and cycloidal tooth profile through cyclic calculation of the pin tooth and cycloidal tooth profile, and then the minimum relative position vector $\mathbf{d}_{ij \min}$ between the point of the cycloidal tooth profile and the center of the pin tooth is obtained, as shown in Fig. 6.

The unit normal vector of the contact line direction for each pin tooth can be expressed as follows:

$$\mathbf{n}_j = \mathbf{d}_{ij \min} / \|\mathbf{d}_{ij \min}\|. \quad (37)$$

In the direction of contact line action, the relative normal velocity of the cycloidal tooth profile and the pin tooth is calculated as follows:

$$v_{n_j} = (\dot{\mathbf{r}}^{Q_i} - \dot{\mathbf{r}}^{P_j})^T \mathbf{n}_j, \quad (38)$$

where $\dot{\mathbf{r}}^{Q_i}$ is the velocity vector at the point of the cycloid profile; $\dot{\mathbf{r}}^{P_j}$ is the velocity vector at the center of the pin; since the pin tooth is assumed to be stationary, there exists $\dot{\mathbf{r}}^{P_j} = [00]^T$.

The following formula can be used to determine the contact force between the cycloidal tooth profile and the pin tooth:

$$F_j = K_{pc}(\delta_{ij \max})^{10/9} + c_c C_{pc} v_{n_j}, \quad (39)$$

where K_{pc} is the contact stiffness of the cycloidal-pin; C_{pc} is the contact damping of the cycloidal-pin; c_c is the damping adjustment factor.

Affected by the change of the radius of curvature of the cycloidal tooth profile, the value of the contact stiffness varies depending on the contact position of the pin tooth and the cycloidal tooth profile. To simplify the calculation, the contact stiffness of cycloidal-pin is set as a constant in the model, and its value is calculated by the mean radius of curvature of the cycloid based on the Hertz contact formula. The detailed calculation of the contact stiffness of the cycloidal pin tooth can be found in the literature [37].

Furthermore, the damping adjustment factor's primary function is to guarantee the contact analysis process's stability. It can be expressed as follows:

$$c_c = \begin{cases} 0 & \delta_{ij \max} < \delta_0 \\ \frac{\delta_{ij \max} - \delta_0}{\delta_1 - \delta_0} & \delta_0 \leq \delta_{ij \max} \leq \delta_1 \\ 1 & \delta_{ij \max} > \delta_1 \end{cases}, \quad (40)$$

where δ_0 and δ_1 are the thresholds for contact depth.

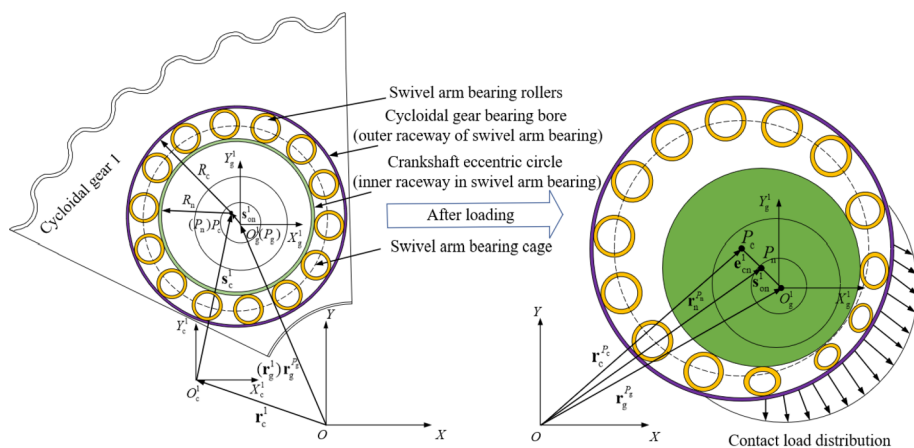


Fig. 7 Contact dynamics modeling of swivel arm bearings

3.5 Contact modeling of swivel arm bearing and support bearing

The swivel arm bearing in the RV-40E reducer is a needle roller bearing, and the supporting bearing is a tapered roller bearing. Only the radial support of the non-standard supporting bearings is taken into account in the modeling process due to its small contact angle. At the same time, the axial preload of the supporting bearings is translated into radial preload in accordance with the contact angle's size. Taking a set of swivel arm bearings and support bearings as an example, the contact modeling of the swivel arm bearings is shown in Fig. 7, and the contact modeling of the support bearings is shown in Fig. 8.

In the contact modeling of the swivel arm bearing, the eccentric outer cylindrical surface of the crankshaft and the inner cylindrical surface of the cycloid bearing hole are regarded as the inner and outer raceways of the bearing rollers. Under the generalized coordinate OXY , the position vectors of the geometrical centers of the inner and outer raceways of the bearing, P_n and P_c , can be expressed as follows:

$$\mathbf{r}_c^{P_c} = \mathbf{r}_c^1 + \mathbf{A}_c^1 \mathbf{s}_c^1, \quad (41)$$

$$\mathbf{r}_n^{P_n} = \mathbf{r}_g^{P_g} + \mathbf{A}_g^1 \mathbf{s}_{on}^1, \quad (42)$$

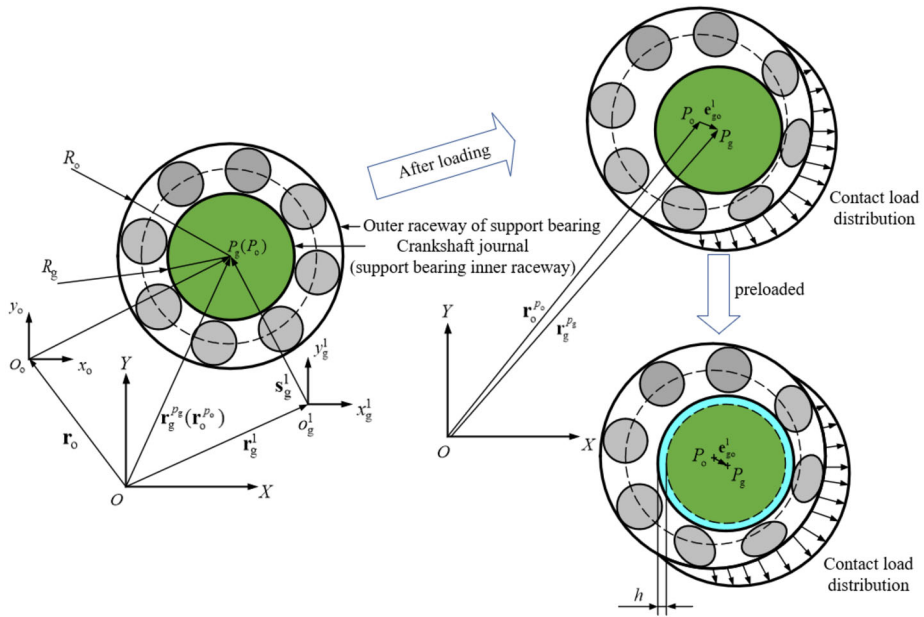
where $\mathbf{r}_g^{P_g}$ is the position vector of the crankshaft rotation center in the generalized coordinate; $\mathbf{s}_c^1$ is the position vector of the geometrical center of the outer raceway of the bearing in the local coordinate of the cycloidal gear 1; $\mathbf{s}_{on}^1$ is the crankshaft eccentricity vector; $\mathbf{A}_c^1$ and $\mathbf{A}_g^1$ are the coordinate transformation matrix.

The relative eccentricity vector between the geometrical centers of the inner and outer raceways after the force is applied to the swivel arm bearing by the effect of clearance and elastic deformation can be expressed as:

$$\mathbf{e}_{cn}^1 = \mathbf{r}_c^{P_c} - \mathbf{r}_n^{P_n}. \quad (43)$$

Equation (43) is derived for time to obtain the relative eccentricity velocity vector:

$$\dot{\mathbf{e}}_{cn}^1 = \dot{\mathbf{r}}_c^{P_c} - \dot{\mathbf{r}}_n^{P_n}. \quad (44)$$



The following formula is used to get the bearing eccentricity distance:

The rolling elements' angular velocity is computed as follows, assuming that they spin synchronously and are evenly distributed inside the bearing ring:

where R_n and R_c are the radius of the inner and outer raceways of the swivel arm bearings respectively; ω_{in} and ω_{out} are the bearings' inner and outer raceways' respective angular velocities, which are determined by the angular velocities of the crankshaft and the cycloidal gear.

where φ_0^k is the initial position angle of each rolling element; k is the serial number of the rolling element of the swivel arm bearing.

where the displacement components of the eccentricity vector $\mathbf{e}_{\text{cn}}$ in the direction of the generalized coordinate's coordinate axes are denoted by the characters e_{cnx} and e_{cny} .

The following is the expression for each rolling element's relative eccentricity velocity in its phase direction:

$$v_b^1 = v_{cnx}^1 \cos \varphi_b^k + v_{cny}^1 \sin \varphi_b^k, \quad (49)$$

where the velocity components of the relative velocity vector $\dot{\mathbf{e}}_{cn}^1$ in the direction of the generalized coordinate axes are denoted by the characters v_{cnx}^1 and v_{cny}^1 .

The normal contact force between each rolling element and the bearing's raceway, ignoring the radial clearance condition, can be represented as follows:

$$F_b^1 = K_b(\delta_b^1)^{10/9} + c_c C_b v_b^1, \quad (50)$$

where K_b and C_b are the rolling element and raceway total contact stiffness and damping, respectively.

The support bearing's contact modeling method is comparable to that of the swivel arm bearing discussed before, the only distinction is that the preload effect must be taken into account, as shown in Fig. 8. The following is an expression for the position vectors of geometrical centers of the support bearing's inner and outer raceways, P_g and P_o , in the generalized coordinate:

$$\mathbf{r}_g^{P_g} = \mathbf{r}_g^1 + \mathbf{A}_g^1 \mathbf{s}_g^1, \quad (51)$$

$$\mathbf{r}_o^{P_o} = \mathbf{r}_o + \mathbf{A}_o \mathbf{s}_o, \quad (52)$$

where $\mathbf{r}_o$ is the position vector of the origin of the local coordinate of the planetary carrier in the generalized coordinate; $\mathbf{s}_o$ is the position vector of the geometric center of the outer raceway of the support bearing in the local coordinate of the planetary carrier; $\mathbf{A}_o$ is the coordinate transformation matrix.

The following is an expression for the relative eccentricity vector between the geometric centers of the support bearing's inner and outer raceways:

$$\mathbf{e}_{go}^1 = \mathbf{r}_g^{P_g} - \mathbf{r}_o^{P_o}. \quad (53)$$

The following formula is used to determine the radial deflections of each rolling element inside the bearing at various positions:

$$\bar{\delta}_b^1 = e_{gox}^1 \cos \bar{\varphi}_b^{\bar{k}} + e_{goy}^1 \sin \bar{\varphi}_b^{\bar{k}}, \quad (54)$$

where the displacement components of the eccentricity vector $\mathbf{e}_{go}^1$ in the direction of the generalized coordinate axis are represented by the variables e_{gox}^1 and e_{goy}^1 ; $\bar{\varphi}_b^{\bar{k}}$ is the rolling elements' respective angular positions; $\bar{k}$ is the serial number of the rolling element of the support bearing.

The normal contact force between each rolling element of the support bearing and the raceway, taking into account the bearing preload effect, is as follows:

$$\bar{F}_b^1 = \bar{K}_b(\bar{\delta}_b^1 + h)^{10/9} + c_c \bar{C}_b \bar{v}_b^1, \quad (55)$$

where h is the impact of axial preload on the radial component of the support bearing; $\bar{K}_b$ and $\bar{C}_b$ represent the total radial contact stiffness and damping between the rolling element and raceway, respectively; $\bar{v}_b^1$ represents the relative eccentricity speed of each rolling element in the position direction.

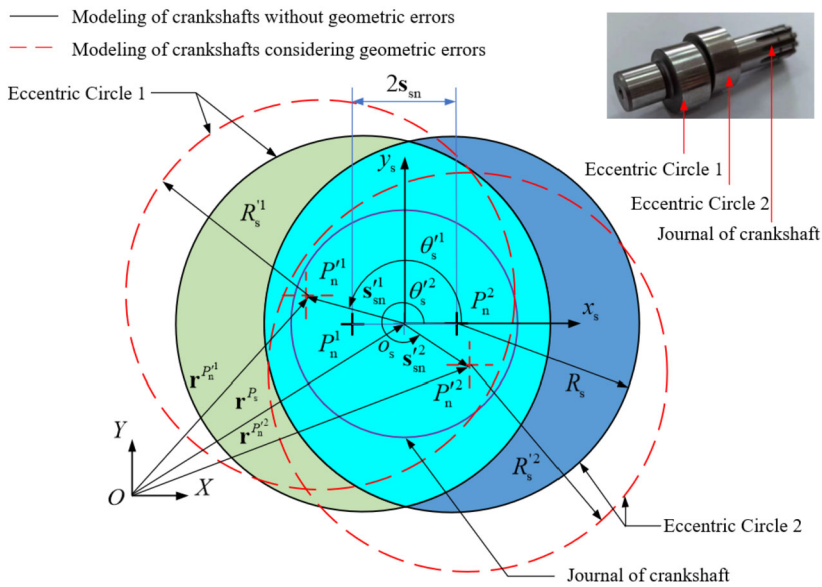


Fig. 9 Modeling of geometric errors in crankshaft

4 Modeling of geometrical errors in critical components of RV reducers for industrial robot joint

4.1 Modeling of geometric errors in crankshaft

As shown in Fig. 9, the main geometric errors of the crankshaft are eccentric pitch length error, eccentric phase error, and eccentric circle diameter error. The centers P_n^1 and P_n^2 of the two eccentric circles on a crankshaft are on the same distribution circle with a 180° phase difference in the ideal model, which does not account for geometric errors. The centers of the two eccentric circles, however, might not be on the same distribution circle as intended due to machining errors, and the angular phase might not be the intended 180° . The geometrical centers of the crankshaft's two eccentric circles are found at $P_n^{\prime 1}$ and $P_n^{\prime 2}$ after taking these geometrical errors into account [15].

The following equation determines the non-ideal positions of the two eccentric centers in the local coordinate $x_s o_s y_s$ of the crankshaft:

$$s_{sn}^{\prime 1}(s_{sn}^{\prime 2}) = s_{sn} + \text{Rand}[l_{sn}^0 \ l_{sn}^1] \quad (56)$$

$$R_s^{\prime 1}(R_s^{\prime 2}) = R_s + \text{Rand}[l_s^0 \ l_s^1] \quad (57)$$

$$\theta_s^{\prime 1} = \pi + \text{Rand}[\phi_s^0 \ \phi_s^1] \quad (58)$$

$$\theta_s^{\prime 2} = 2\pi + \text{Rand}[\phi_s^0 \ \phi_s^1] \quad (59)$$

where, considering geometric errors, $s_{sn}^{\prime 1}$ and $s_{sn}^{\prime 2}$ are the eccentric distances from the crankshaft eccentric circle center to the local coordinate origin. s_{sn} is the crankshaft's ideal eccentricity when geometrical errors are ignored; $\text{Rand}[l_{sn}^0 \ l_{sn}^1]$ means that a value is randomly selected within the length error range $[l_{sn}^0 \ l_{sn}^1]$; $R_s^{\prime 1}$ and $R_s^{\prime 2}$ are the actual eccentric

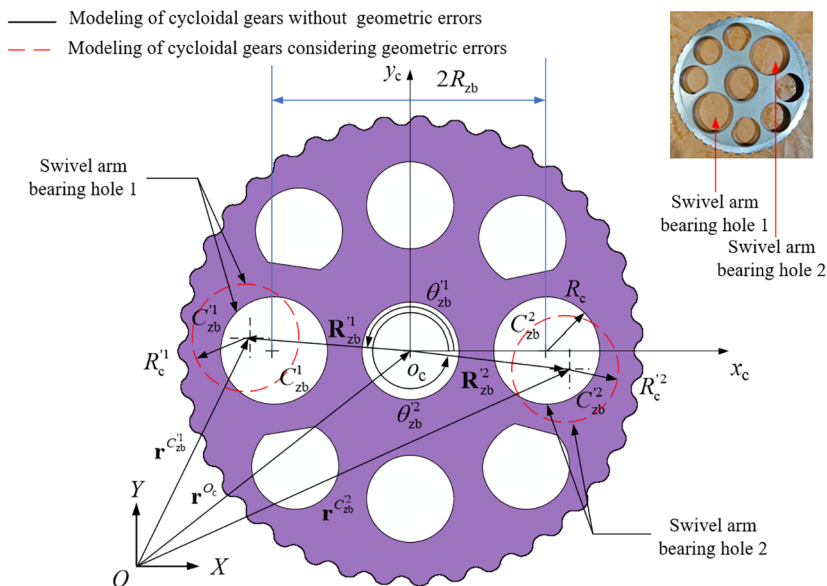


Fig. 10 Modeling of geometric errors in a cycloidal gear

radius of the crankshaft considering geometrical errors; $\theta_s'^1$ and $\theta_s'^2$ are the actual eccentricity phase angles of the crankshaft considering geometrical errors; $\text{Rand}[\phi_s^0 \ \phi_s^1]$ means that a value is randomly selected within the angular error range $[\phi_s^0 \ \phi_s^1]$; $[l_s^0 \ l_s^1]$ is the range of crankshaft eccentricity diameter error; $\text{Rand}[] = \text{Lowerdeviation} + (\text{Upperdeviation} - \text{Lowerdeviation}) \cdot \text{Rand}()$ and $\text{Rand}()$ indicates a random value between 0 and 1.

In the generalized coordinate, the location vectors of the two eccentric circle centers on the crankshaft, as indicated in Fig. 9, can be represented as follows after taking the geometric error into account:

$$\mathbf{r}_{n'}^{p/1} = \mathbf{r}^p + \mathbf{A}_s \mathbf{s}_{sn}'^1, \quad (60)$$

$$\mathbf{r}_{n'}^{p/2} = \mathbf{r}^p + \mathbf{A}_s \mathbf{s}_{sn}'^2, \quad (61)$$

where $\mathbf{A}_s$ is the coordinate transformation matrix.

4.2 Modeling of geometric errors in cycloidal gears

The main geometric errors of the cycloidal gears include the phase angle error of the bearing hole center distribution, the radius error of the bearing hole center distribution, the bearing hole diameter error, and the geometric error of the cycloidal tooth shape machining as shown in Fig. 10. In the model without geometrical errors, the centers of the swivel arm bearing bores C_{zb}^1 and C_{zb}^2 of the cycloidal gear are located on the same distribution circle with a phase difference of 180° . However, due to machining errors, the centers of the two swivel arm bearing bores are not necessarily on the same distribution circle, and the angular phase is not always the desired 180° . After accounting for these geometric errors, the two swivel arm bearing bores are centered at the non-ideal points $C_{zb}'^1$ and $C_{zb}'^2$.

In the local coordinate $x_c o_c y_c$ of the cycloidal gears, the non-ideal position of the centers of the two swivel arm bearing bores is determined by the following equations:

$$R'_{zb}(R'^2_{zb}) = R_{zb} + \text{Rand}[l^0_{zb} \ l^1_{zb}], \quad (62)$$

$$R'_c(R'^2_c) = R_c + \text{Rand}[l^0_c \ l^1_c], \quad (63)$$

$$\theta'^1_{zb} = \pi + \text{Rand}[\phi^0_{zb} \ \phi^1_{zb}], \quad (64)$$

$$\theta'^2_{zb} = 2\pi + \text{Rand}[\phi^0_{zb} \ \phi^1_{zb}], \quad (65)$$

where R'^1_{zb} and R'^2_{zb} are the radial distances from the center of the swivel arm bearing hole to the origin of the local coordinate $x_c o_c y_c$, considering geometric errors. R_{zb} is the radial distance from the center of the swivel arm bearing bore to the center of the cycloidal gear without considering geometric errors; R'^1_c and R'^2_c indicate the actual radius of the bearing bores of the cycloidal gear considering geometrical errors; θ'^1_{zb} and θ'^2_{zb} are phase angles distributed in the center of the swivel arm bearing hole considering geometric errors; $[l^0_{zb} \ l^1_{zb}]$ is the range of cycloidal gears bearing hole center distribution radius error; $[l^0_c \ l^1_c]$ is the range of cycloidal gears bearing hole diameter error; $[\phi^0_{zb} \ \phi^1_{zb}]$ is the range of cycloidal gears bearing hole center distribution phase angle error.

As shown in Fig. 10, after considering the geometric error, the position vector of the center of the swivel arm bearing hole on the cycloidal gears in the generalized coordinate can be rewritten as:

$$\mathbf{r}^{C'1}_{zb} = \mathbf{r}^{O_c} + \mathbf{A}_c \mathbf{R}^1_{zb}, \quad (66)$$

$$\mathbf{r}^{C'2}_{zb} = \mathbf{r}^{O_c} + \mathbf{A}_c \mathbf{R}^2_{zb}. \quad (67)$$

In addition to the above geometric error, the cycloidal gear tooth shape processing error is also an important factor affecting the transmission accuracy of the RV reducer. Affected by the randomness of tooth shape processing error, the actual shape of each tooth of the cycloidal gear will not be exactly the same, which makes it difficult to express the actual tooth shape error. As shown in Fig. 11, the cycloidal gear tooth shape machining error is generally tested by three coordinate measuring machine, and the range of cycloidal tooth shape deviation is obtained by comparing the actual tooth shape curve with the theoretically designed tooth shape curve, taking the maximum outer deviation value and the maximum inner deviation value as the reference and defining it as the cycloidal tooth profile tooth shape error. In the figure, the two dotted lines are the equidistant curves of the theoretical tooth profile of the pendulum line obtained based on the maximum external deviation value and the maximum internal deviation value. Taking the equidistant curve as a reference, the equidistant trimmed tooth profile will be used to simulate the cycloidal tooth profile machining error in the modeling of the cycloidal tooth profile error.

4.3 Modeling of geometric errors in planetary carrier

The main geometric errors of the planetary carrier include the center-distributed phase angle error of the support bearing holes and the center-distributed radial length error, as shown in Fig. 12. In the model without geometrical errors, the centers O^1_{zc} and O^2_{zc} of the output planetary carrier support bearing holes are located on the same distribution circle with a phase difference of 180° . However, due to machining errors, the centers of the two support

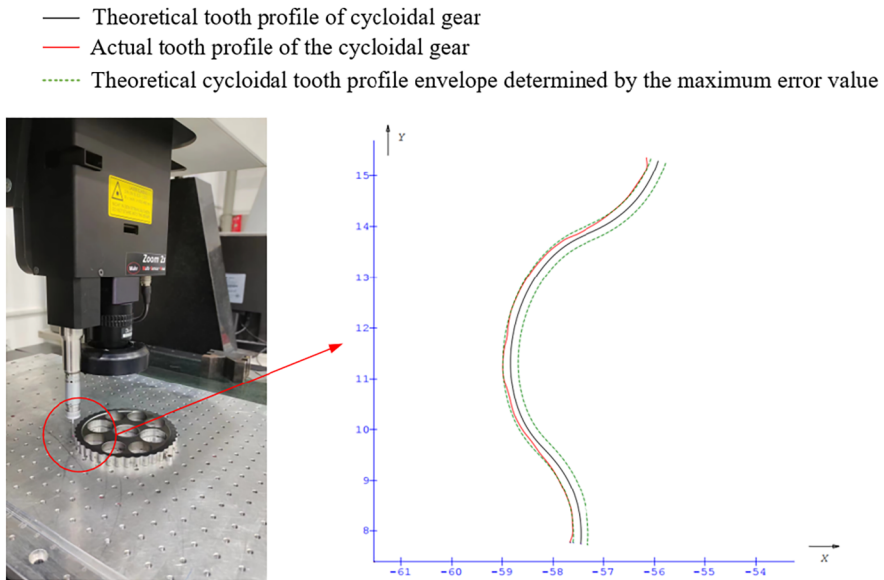


Fig. 11 RV reducer cycloidal tooth shape inspection

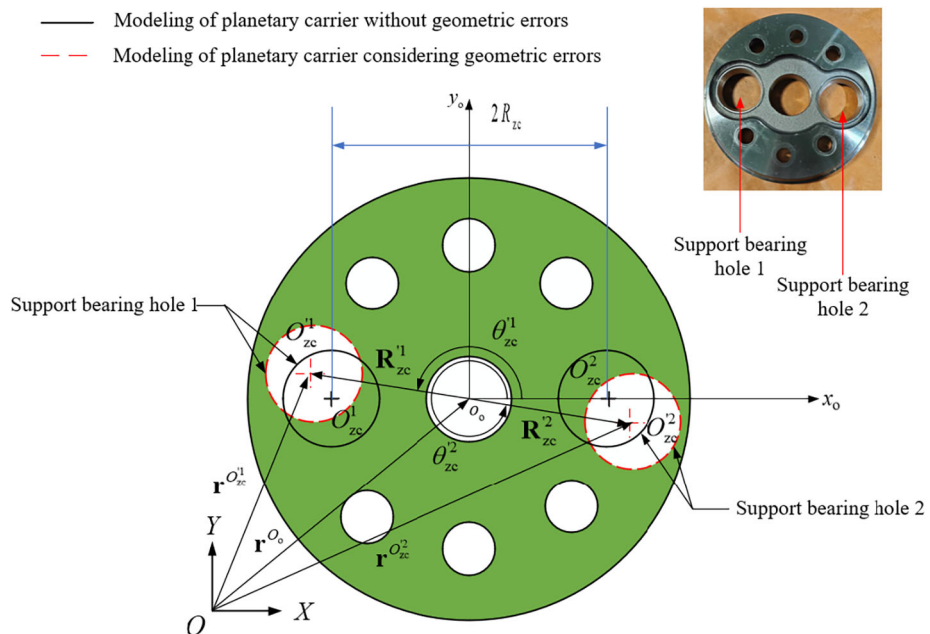


Fig. 12 Modeling of geometric errors in planetary carrier

bearing bores are not necessarily on the same distribution circle, and the phase difference is not always the desired 180° . After accounting for these geometric errors, the two support bearing bores are centered at the non-ideal points O_{zc}^1 and O_{zc}^2 .

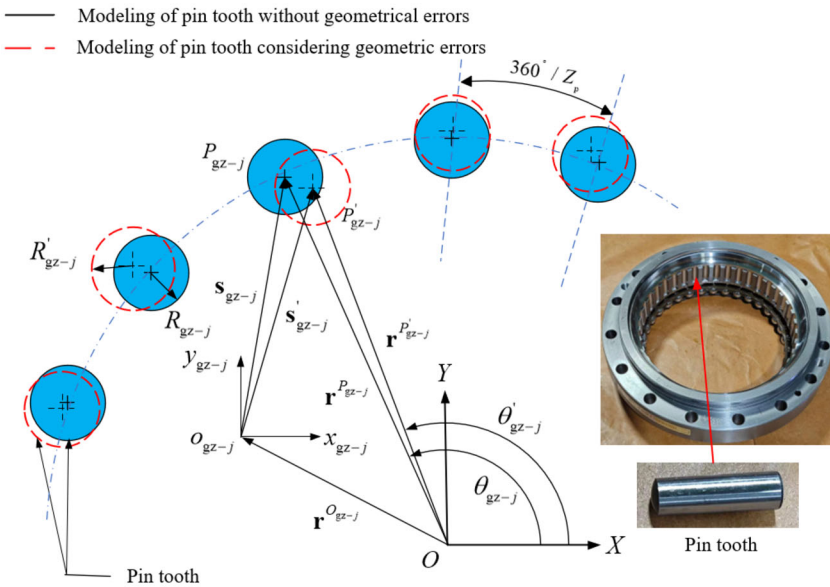


Fig. 13 Modeling of geometric errors in pin tooth

In the local coordinate $x_o o_o y_o$ of the planetary carrier, the non-ideal position of the centers of the two support bearing bores is determined by the following equations:

$$R'_{zc}(R'^2_{zc}) = R_{zc} + \text{Rand}[l^0_{zc} \ l^1_{zc}], \quad (68)$$

$$\theta'^1_{zc} = \pi + \text{Rand}[\phi^0_{zc} \ \phi^1_{zc}], \quad (69)$$

$$\theta'^2_{zc} = 2\pi + \text{Rand}[\phi^0_{zc} \ \phi^1_{zc}], \quad (70)$$

where R'^1_{zc} and R'^2_{zc} are the radial distances from the center of the support bearing hole relative to the origin of the local coordinate $x_o o_o y_o$ considering geometrical errors; R_{zc} is the radial distance from the center of the support bearing hole to the center of the output planetary carrier without considering geometrical errors; θ'^1_{zc} and θ'^2_{zc} are the phase angles of the center of the support bearing hole considering geometrical errors; $[l^0_{zc} \ l^1_{zc}]$ is the range of distributed radial length error of planetary carrier support bearing hole center; $[\phi^0_{zc} \ \phi^1_{zc}]$ is the range of phase angle error of planetary carrier support bearing hole center distribution.

As shown in Fig. 12, after considering the geometric error, the position vector of the center of the support bearing hole on the output planetary carrier in the generalized coordinate can be rewritten as

$$\mathbf{r}^{O'1}_{zc} = \mathbf{r}^{O_o} + \mathbf{A}_o \mathbf{R}'^1_{zc}, \quad (71)$$

$$\mathbf{r}^{O'2}_{zc} = \mathbf{r}^{O_o} + \mathbf{A}_o \mathbf{R}'^2_{zc}. \quad (72)$$

4.4 Modeling of geometric errors in pin tooth

As shown in Fig. 13, the main geometric errors of pin tooth are each pin tooth diameter error, each pin tooth distribution radius error and each pin tooth distribution angle error. While it

is ideal for all pin diameters to be the same and for each pin tooth's geometric center to be evenly spaced on a single circle, geometric errors can cause individual pin diameters to vary and the geometric centers of pin teeth to not always be evenly spaced on a single circle [15].

The formula is as follows:

$$R'_{gz-j} = R_{gz-j} + \text{Rand}[l_{R_{gz}}^0 \ l_{R_{gz}}^1], \quad (73)$$

$$s'_{gz-j} = r_p + \text{Rand}[l_{s_{gz}}^0 \ l_{s_{gz}}^1], \quad (74)$$

$$\theta'_{gz-j} = \theta_{gz-j} + \text{Rand}[\phi_{gz}^0 \ \phi_{gz}^1], \quad (75)$$

where R'_{gz-j} is the actual radius of the j pin tooth after considering the geometric error; R_{gz-j} is the theoretical radius of the j pin tooth; s'_{gz-j} is the actual radial distance of the j pin tooth center in its local coordinate after considering geometric errors; r_p is the radius of the ideal distribution circle of the pin tooth; θ'_{gz-j} is the actual distributed phase angle in the local coordinate for the j pin tooth after considering geometric errors; θ_{gz-j} is the theoretical distributed phase angle in the local coordinate for the j pin tooth; $[l_{R_{gz}}^0 \ l_{R_{gz}}^1]$ is the range of pin tooth diameter error; $[l_{s_{gz}}^0 \ l_{s_{gz}}^1]$ is the range of pin tooth distribution radius error; $[\phi_{gz}^0 \ \phi_{gz}^1]$ is the range of pin tooth distribution angle error.

It is possible to modify the location of each pin tooth's center by using Eq. (74) and Eq. (75). The position vector of the geometrical center of the j pin tooth considering the geometrical errors in the generalized coordinate can be expressed as follows:

$$\mathbf{r}^{P'_{gz-j}} = \mathbf{r}^{O_{gz-j}} + \mathbf{A}_{gz-j} \mathbf{s}'_{gz-j}. \quad (76)$$

4.5 Modeling of geometric errors in swivel arm bearings

The radial clearance of the swivel arm bearing is the main factor affecting the transmission accuracy of the RV reducer, and the size of the radial clearance is determined by the diameter of the inner and outer raceways of the bearing as well as the dimensional error of the diameter of the rolling element. As shown in Fig. 14, the main geometric errors of the swivel arm bearing are an inner raceway diameter error, an outer raceway diameter error, and a rolling element diameter error. Since there is no separate inner and outer ring for the swivel arm bearing in the RV reducer, the diameter error of the inner raceway of the swivel arm bearing is the diameter error of the eccentric circle of the crankshaft, and the diameter error of the outer raceway of the swivel arm bearing is the diameter error of the hole of the cycloidal gears bearing. The method of expressing the inner and outer raceway errors of the swivel arm bearings has been described in detail in the crankshaft and cycloidal gear geometry error modeling.

The rolling element radius error is expressed as follows:

$$R'_{gdt-i} = R_{gdt-i} + \text{Rand}[l_{gdt}^0 \ l_{gdt}^1], \quad (77)$$

where R'_{gdt-i} is the actual radius of the i rolling element of the swivel arm bearing considering the geometric error; $[l_{gdt}^0 \ l_{gdt}^1]$ is the range of swivel arm bearing rolling element diameter error.

In addition, since tapered roller bearings and angular contact ball bearings are preloaded, the effect of the rotary error of these two types of bearings is ignored.

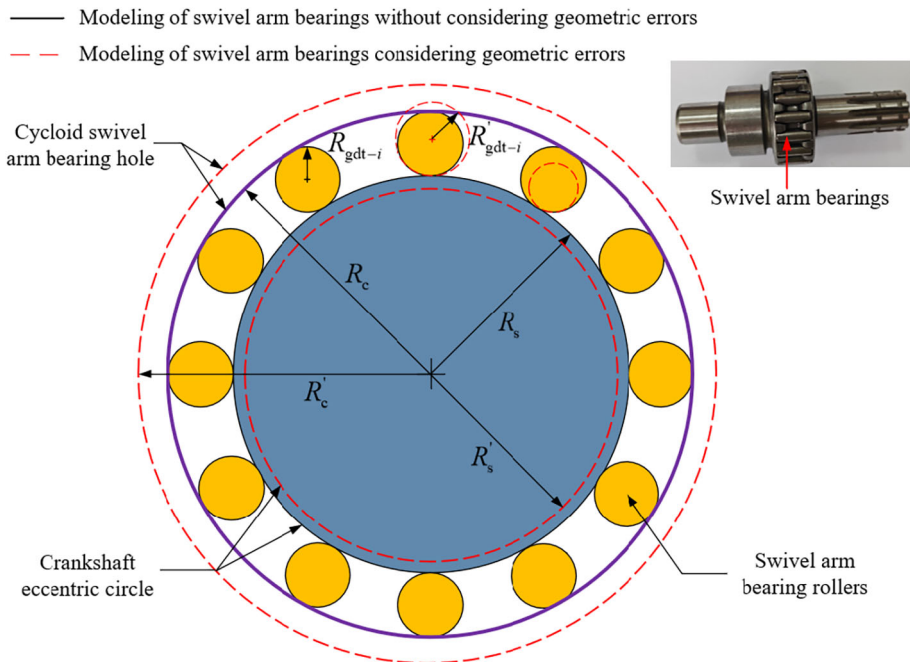


Fig. 14 Modeling of geometric errors in swivel arm bearings

5 Model validation of the RV reducer for industrial robot joint

Reducer transmission error test is devised through the front and rear of the two grating angle encoders to collect input and output angle, with the output angle minus the input angle divided by the transmission ratio to get the transmission error of the reducer, as shown in Fig. 15. However, the RV reducer is a precision drive, and current optical angle encoders can only capture data at constant and low speeds, not at variable speeds, so this study verifies the correctness of the theoretical dynamic model through the transmission error test of the RV reducer under equal-speed conditions, and then carries out the analysis of the dynamic transmission accuracy of the RV reducer under variable-speed conditions based on the theoretical model. The main design parameters of the RV-40E reducer are shown in Table 1, the dynamic parameters of the RV-40E reducer used in simulation are shown in Table 2, the geometrical error parameters of critical transmission components of the RV-40E reducer are shown in Table 3.

The fluctuation of the output angle of the planetary carrier and the input angle of the sun gear determines the transmission error of the RV reducer. The input and output rotational angle data of the RV reducer can be obtained through a dynamic analysis on the multibody dynamics model of the RV-40E reduction, as previously mentioned. Following is the calculation for the transmission error:

$$\varepsilon = \frac{\varphi_s}{i_{s-p}} - \varphi_0, \quad (78)$$

where i_{s-p} is the ratio of the RV-40E reducer, φ_s is the input angle of the sun gear, and φ_0 is the output angle of the planetary carrier.

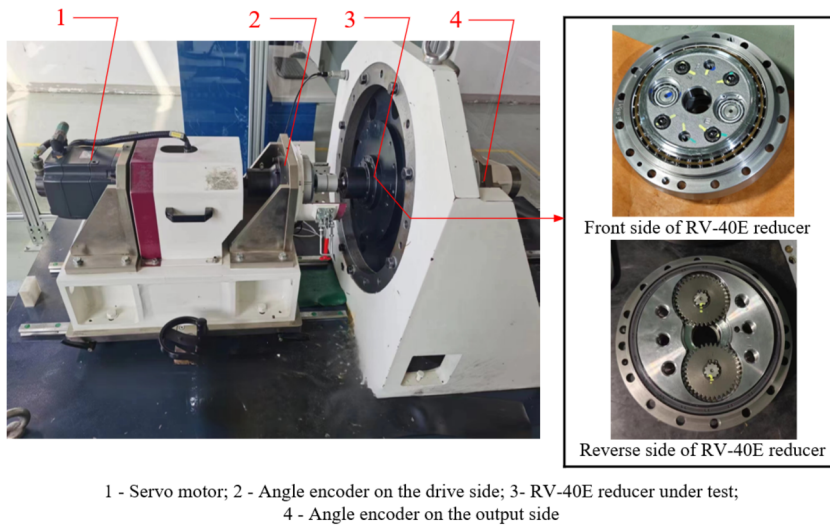


Fig. 15 Reducer transmission error test equipment

Table 1 The main design parameters of the RV-40E reducer

Parameters	Values
Number of sun gear teeth	12
Number of planetary gear teeth	36
Number of pins	40
Number of teeth of cycloidal gears	39
Gear ratio i_{s-p}	121
Radius of the center circle of the pin gear (mm)	66
Crankshaft eccentricity (mm)	1.3
Pin tooth radius (mm)	3

The findings of the frequency domain analysis of the transmission error can be acquired by performing a fast Fourier transform (FFT) on the time domain signal of the transmission error. f stands for the frequency of transmission error. f_o is the planetary carrier's rotation frequency, which can be found by dividing the sun gear's rotation frequency by the RV reducer's transmission ratio. The number of transmission error variations per revolution of the output planetary carrier is represented by the frequency ratio, f/f_o .

The theoretical model of the RV-40E reducer's time-domain response signal for transmission error is displayed in Fig. 16(a). The time-domain signal of the transmission error fluctuates between $(0.483', -0.412')$, which means that the reducer's transmission error is $0.895'$. The transmission error fulfills the requirements of industrial robot joints for the transmission accuracy performance of the RV reducer. Figure 16(b) shows the experimental time-domain response signal of the transmission error of the RV-40E reducer. The fluctuation range of its transmission error time-domain signal is located at $(0.35', -0.47')$, which means that the transmission error of the RV reducer is $0.82'$.

Figure 17 shows the results of the frequency domain analysis, where the horizontal axis coordinate is the frequency ratio. It can be seen that both the theoretical analysis results

Table 2 The dynamic parameters of RV-40E reducer used in simulation

Parameters	Values
Mass of the input shaft, m_1 (kg)	0.44
Mass of two crankshaft assemblies, m_2, m_3 (kg)	0.25
Mass of two-piece cycloidal gears, m_4, m_5 (kg)	0.45
Mass of output planetary carrier, m_6 (kg)	3.379
Inertia of the input shaft, I_1 (kg mm ²)	70
Inertia of two crankshaft assemblies, I_2, I_3 (kg mm ²)	48
Inertia of two-piece cycloidal gears, I_4, I_5 (kg mm ²)	1100
Inertia of output planetary carrier, I_6 (kg mm ²)	7600
Modulus of elasticity, E (GPa)	207
Poisson's ratio, ν	0.3
Contact stiffness of cycloid and pin tooth, K_{pc} (N/m)	0.7×10^8
Contact stiffness of gear pair, K_t (N/m)	6.639×10^7
Contact stiffness of swivel arm bearings, K_b (N/m)	0.7×10^8
Normal contact stiffness of support bearing, $\bar{K}_b$ (N/m)	1.5×10^8
Contact damping, $C_{pc}, C_t, C_b, \bar{C}_b$ (N s/m)	500
Gravitational acceleration, g (m/s ²)	9.8
Input rotational speed (rpm)	1500

Table 3 The geometrical error parameters of critical transmission components of the RV-40E reducer

Critical components	Geometric error term	Lower deviation	Upper deviation
Crankshaft	Crankshaft eccentricity length error (μm)	$l_{sn}^0 = -3$	$l_{sn}^1 = +3$
	Crankshaft eccentricity angle error (")	$\phi_s^0 = -20$	$\phi_s^1 = +20$
Cycloidal gears	Cycloidal gears bearing hole center distribution phase angle error (")	$\phi_{zb}^0 = -20$	$\phi_{zb}^1 = +20$
	Cycloidal gears bearing hole center distribution radius error (μm)	$l_{zb}^0 = -3$	$l_{zb}^1 = +3$
Planetary carrier	Phase angle error of planetary carrier support bearing hole center distribution (")	$\phi_{zc}^0 = -20$	$\phi_{zc}^1 = +20$
	Distributed radial length error of planetary carrier support bearing hole center (μm)	$l_{zc}^0 = -3$	$l_{zc}^1 = +3$
Pin tooth	Pin tooth distribution angle error (")	$\phi_{gz}^0 = -40$	$\phi_{gz}^1 = +40$
	Pin tooth distribution radius error (μm)	$l_{sgz}^0 = 0$	$l_{sgz}^1 = +3$
	Pin tooth diameter error (μm)	$l_{Rgz}^0 = -3$	$l_{Rgz}^1 = -1$
Swivel arm bearings	Swivel arm bearing rolling element diameter error (μm)	$l_{gdt}^0 = -3$	$l_{gdt}^1 = -3$
	Tolerance of inner raceway diameter of swivel arm bearing (crankshaft eccentricity diameter error) (μm)	$l_s^0 = -3$	$l_s^1 = -1$
	Tolerance of outer raceway diameter of swivel arm bearing (cycloidal gears bearing hole diameter error) (μm)	$l_c^0 = +1$	$l_c^1 = +3$

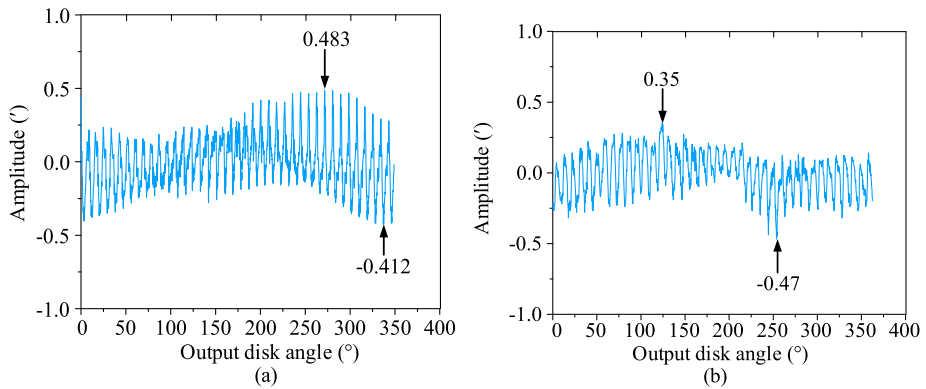


Fig. 16 Time-domain signals of transmission errors under equal speed and no-load conditions: (a) theoretical analysis results; (b) test analysis results

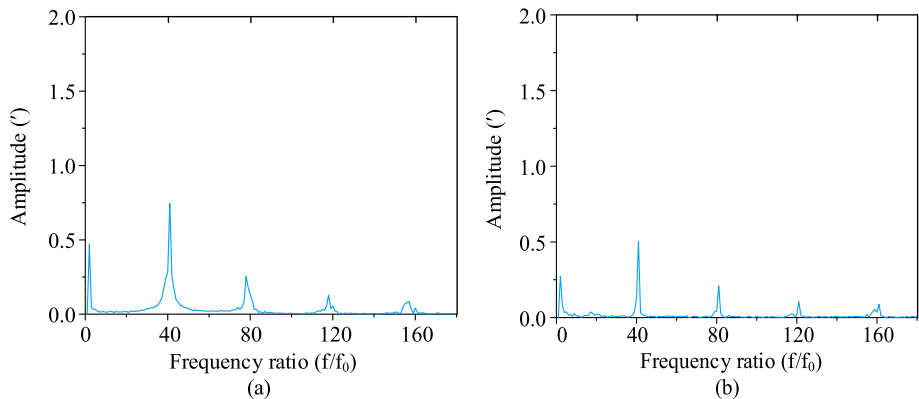
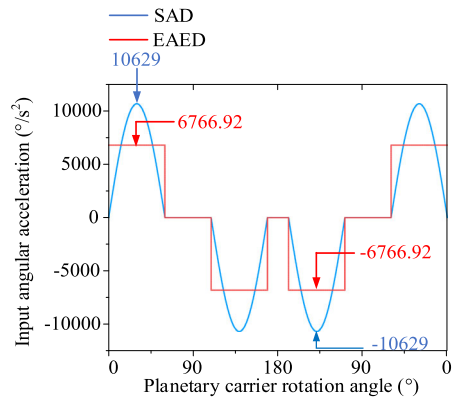
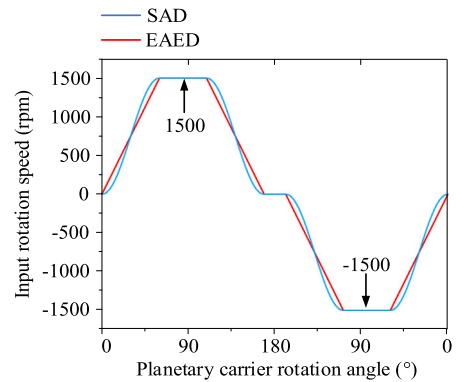


Fig. 17 Frequency-domain signals of transmission errors under equal speed and no-load conditions: (a) theoretical analysis results; (b) test analysis results

and the experimental results, the frequency components of the transmission error of the RV reducer are mainly concentrated in the frequencies of 40, 80, 120, and 160, which are reflected in the multiples of the engagement frequency of the cycloid transmission.

By comparing the theoretical simulation results and test results of the transmission error of the RV-40E reducer, it can be learned that the amplitude of the time domain response of the transmission error of the RV reducer is relatively close to each other, and all of them satisfy the requirement of accuracy less than $1'$, which indicates that the theoretical analysis results correspond well with the test data. The regularity of the results of frequency domain analysis of the transmission error of the RV reducer is also consistent, and the main frequency components are 40, 80, 120, 160, etc., and the most significant values all appear at 40, indicating that the cycloidal-pin transmission has a greater impact on the transmission error. Therefore, the accuracy of the theoretical model can be fully verified, whether through the comparison of time domain or frequency domain signals.

Fig. 18 Input angular acceleration**Fig. 19** Input rotational speed

6 Transmission error analysis of RV reducer for industrial robot joint under variable operating conditions

6.1 Dynamic transmission error analysis of the RV reducer without considering the effects of load inertia

Based on the two typical driving laws EAED and SAD established in Chap. 2, the effects of different driving laws on the dynamic performance of RV reducers are analyzed under the working condition, without considering the load inertia. Moreover, the analysis was performed by simulating the industrial robot joints oscillating back and forth from 0° to 180° for one cycle. As shown in Fig. 18–Fig. 21, the comparisons of input angular acceleration, input rotational speed, output angular acceleration, and output rotational speed for the same time and rotation angle of the two driving laws are demonstrated, respectively.

With the data in Fig. 20, we can clearly see that the angular acceleration under the SAD driving law is significantly smaller than that under the EAED driving law during the start-stop phase of the RV reducer. The reason for this phenomenon is that there is a step phenomenon in the angular acceleration value of the EAED drive law during the start-stop phase, as shown in Fig. 18. This step phenomenon leads to a more pronounced vibration shock in the RV reducer during this phase. In contrast, the SAD driving law has a smoother change in angular acceleration, as shown in Fig. 18. This smooth transition results in relatively small vibration shocks in the RV reducer during the start-stop phase.

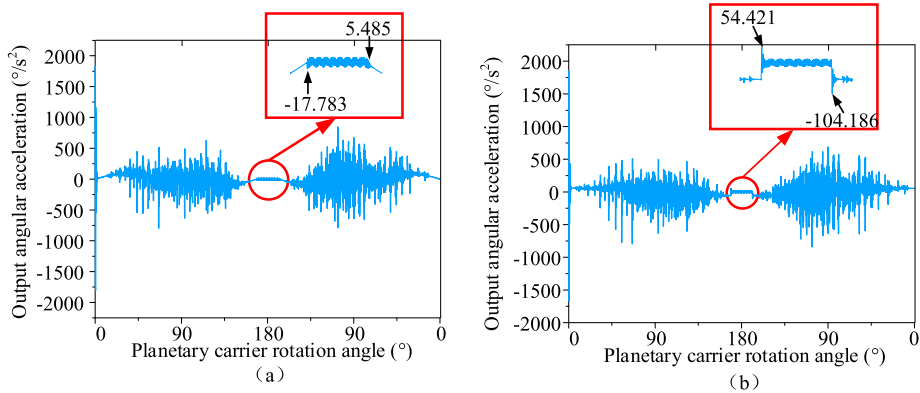


Fig. 20 Output angular acceleration: (a) SAD; (b) EAED

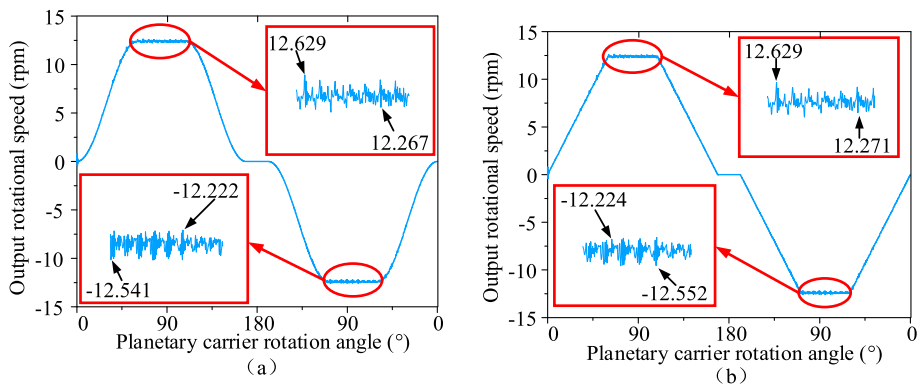


Fig. 21 Output rotational speed: (a) SAD; (b) EAED

Under the working condition without load inertia, the variation range of transmission error of the RV reducer driven by the SAD driving law, the EAED driving law, and the equal speed driving law is shown in Fig. 22. According to the data in Fig. 22, it can be obtained that under the SAD driving law, the fluctuation range of the transmission error of the RV reducer is $(0.323', -0.628')$, which is calculated as $0.951'$; under the EAED driving law, the fluctuation range of the transmission error is $(0.316', -0.624')$, which corresponds to the transmission error of $0.94'$. Compared with the SAD driving law, the error value is slightly reduced, but the difference between the two is not significant; under the equal speed driving law, the fluctuation range of the transmission error is $(0.483', -0.412')$, and the transmission error is calculated to be $0.895'$, as shown in Fig. 16(a). Compared to the equal speed drive law, the amplitude of the transmission error increases by 6.3% for the SAD drive and by 5.1% for the EAED drive.

Under the working condition, without considering the load inertia, different driving laws have a certain effect on the vibration impact of the RV reducer, but do not have a great effect on the transmission error value. Moreover, the results of the analysis in this part can further verify the correctness of the above constructed mathematical model of the driving law and the theoretical model of the RV reducer. However, in the actual working conditions

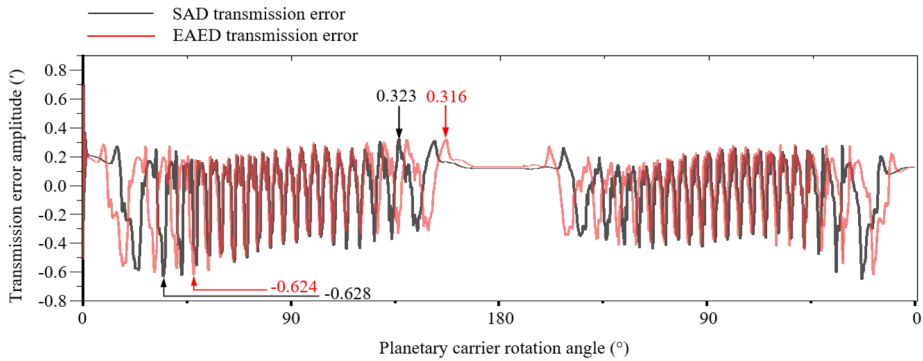


Fig. 22 Transmission errors of RV reducers with different drive laws without load inertia

The equivalent load inertia is $14.35 \text{ kg} \cdot \text{m}^2$ when the weight are 5 kg

The equivalent load inertia is $18.397 \text{ kg} \cdot \text{m}^2$ when the weight are 7.5 kg

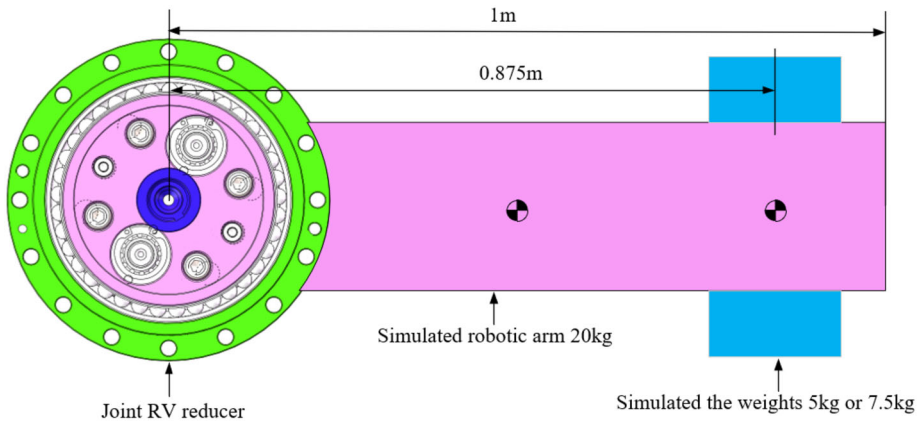
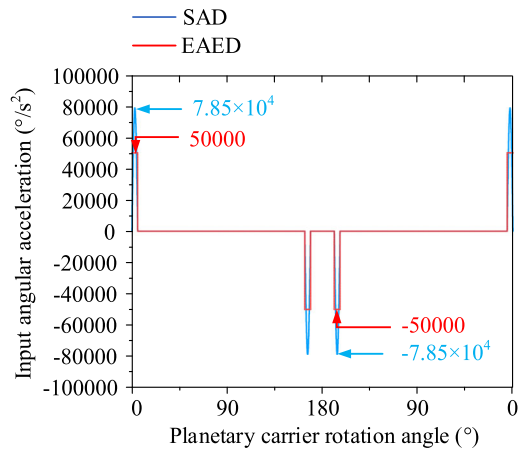
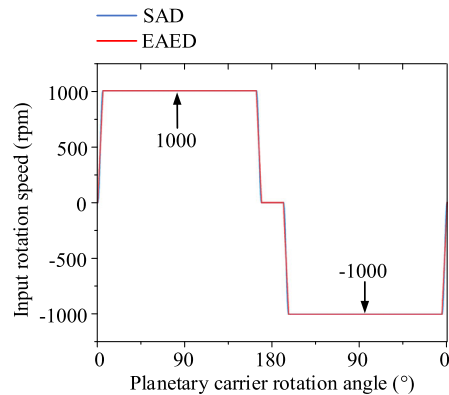


Fig. 23 Model of a simplified mechanical arm

of industrial robot joints, it is still necessary to consider the influence of the load inertia generated by the length and weight of the robotic arm on the joint RV reducer.

6.2 Dynamic transmission error analysis of the RV reducer with considering the effects of load inertia

The effects of two typical drive laws on the dynamic performance of the RV reducer based on the working conditions with a maximum rotational speed of 1000 rpm and a load inertia of $14.35 \text{ kg} \cdot \text{m}^2$ are analyzed. Here, the load inertia $14.35 \text{ kg} \cdot \text{m}^2$ is equated to the rotational inertia of the output planetary carrier of the jointed RV reducer by a simple mechanical arm as shown in Fig. 23. As shown in Fig. 24–Fig. 27, the comparisons of input angular acceleration, input rotational speed, output angular acceleration, and output rotational speed for the same time and rotation angle for the two driving laws are shown, respectively. Through the data in the figure, we can obviously get that the vibration response generated by using the

Fig. 24 Input angular acceleration**Fig. 25** Input rotational speed

SAD driving law is obviously smaller than the vibration response under the EAED driving law, and the reason for the generation is consistent with the aforementioned no-load inertia.

Figure 28 shows the range of transmission errors when the RV reducer is driven by SAD, EAED, and equal speed under the working condition of maximum rotational speed of 1000 rpm and load inertia of 14.35 kg m^2 respectively. According to the data in Fig. 28, it can be obtained that under the SAD driving law, the transmission error of the RV reducer fluctuates in the range of $(4.388', -3.820')$, which is calculated to be $8.208'$. Under the EAED driving law, the fluctuation range of the transmission error is $(5.177', -4.489')$, which corresponds to a transmission error of $9.666'$. Compared with the SAD driving law, the error value increases by $1.458'$, which mainly stems from the sudden change of angular acceleration during EAED, resulting in a significant increase of the transmission error. This reflects that in terms of control accuracy, the SAD driving law is significantly better than the EAED driving law; the fluctuation range of the transmission error under the equal speed driving law is $(0.686', -0.695')$, which is calculated to be $1.381'$. Compared to the equal speed drive law, the amplitude of the transmission error increases by 494.3% for the SAD drive and by 599.9% for the EAED drive. This indicates that the transmission performance of the RV reducer is the most stable and the transmission error is the smallest in the process of equal speed driving law. And it can be obtained from Fig. 28 that after the speed of the variable speed driving law is stabilized, the transmission error of the variable speed drive

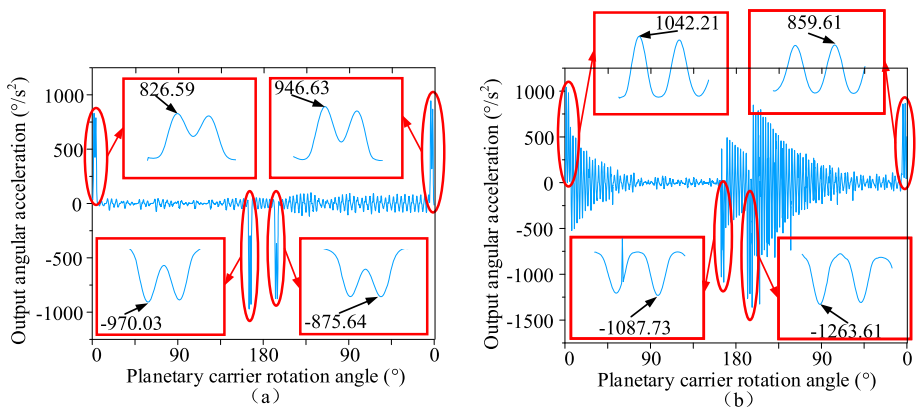


Fig. 26 Output angular acceleration: (a) SAD; (b) EAED

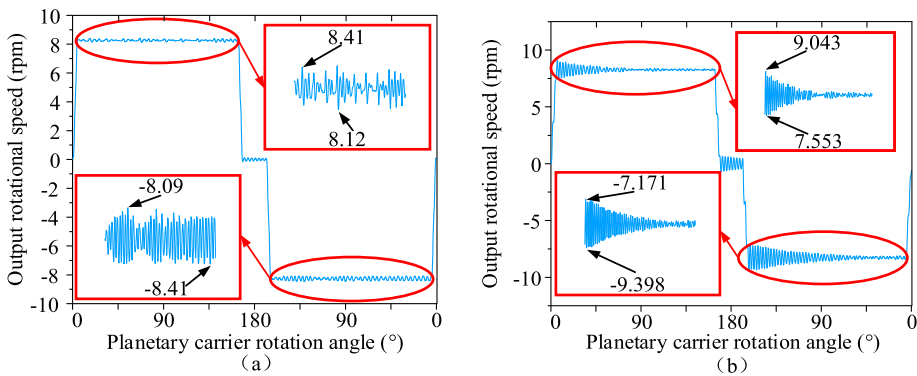


Fig. 27 Output rotational speed: (a) SAD; (b) EAED

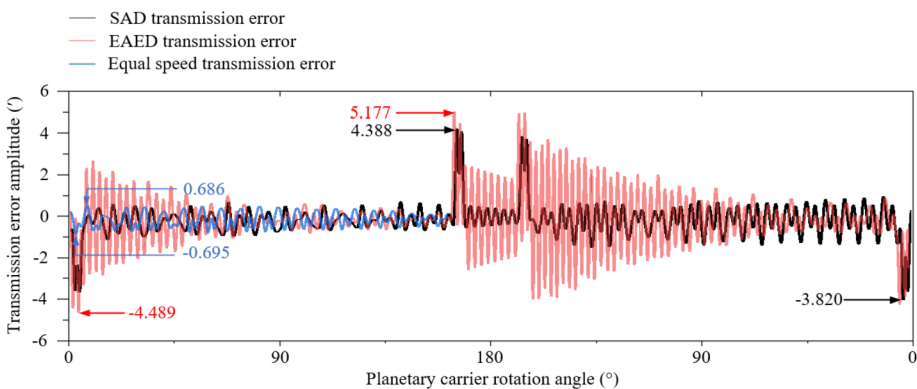


Fig. 28 Comparison of transmission error based on maximum rotational speed of 1000 rpm and load inertia of 14.35 kg m^2

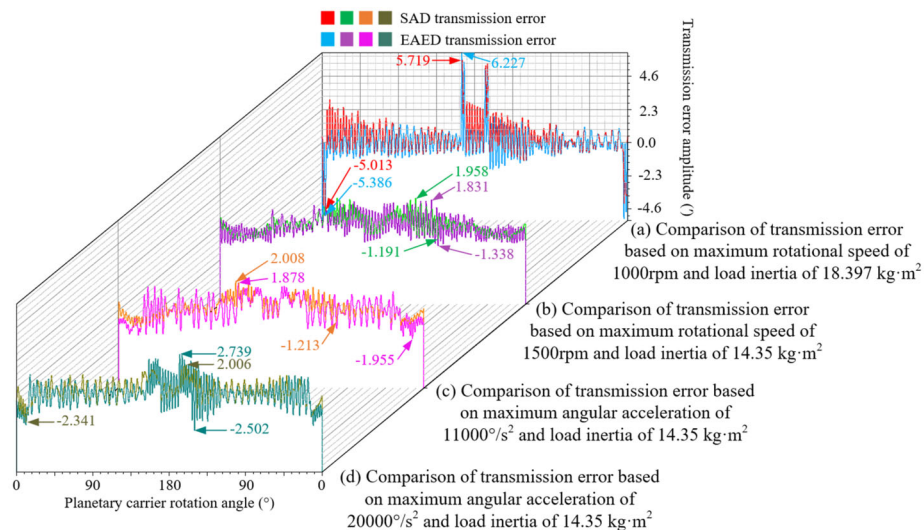


Fig. 29 Transmission error with two variable speed drive laws under different operating conditions

law has approximately the same range of variation as the transmission error of the equal speed drive law, which indicates that the variable speed process is the main reason for the transmission error value of the RV reducer to become larger. However, the existing studies on transmission error are carried out based on the equal speed working condition, ignoring the influence of transmission error generated in the variable-speed process on the joint accuracy of industrial robots, which leads to reduced accuracy in analyzing motion precision of industrial robots.

Figure 29(a) represents the range of transmission error variation of the RV reducer with the SAD driving law and the EAED driving law respectively at the maximum rotational speed of 1000 rpm and load inertia of 18.397 kg m². The fluctuation range of transmission error of the RV reducer under the SAD driving law is (5.719', -5.013'), and the transmission error is calculated to be 10.732'; under the EAED driving law, the fluctuation range of transmission error is (6.227', -5.386'), and the corresponding transmission error is 11.613'. Compared with the SAD driving law, the error value increases by 0.881'. When the rotational speed is the same and the load inertia increases by 28.2%, the transmission error for the SAD drive law increases from 8.208' to 10.732', for the EAED drive law increases from 9.666' to 11.613'.

Figure 29(b) represents the range of transmission error variation of the RV reducer with the SAD driving law and the EAED driving law respectively at the maximum rotational speed of 1500 rpm and load inertia of 14.35 kg m². According to the data in Fig. 25, it can be obtained that under the SAD driving law, the fluctuation range of transmission error of the RV reducer is (1.958', -1.191'), and the transmission error is calculated to be 3.149'; under the EAED driving law, the fluctuation range of transmission error is (1.831', -1.338'), and the corresponding transmission error is 3.169'. Compared with the SAD driving law, the error value increases by 0.02'. When the load inertia is the same and the rotational speed is increased by 50%, the transmission errors of the two drive laws are reduced by 61.6% and 67.2%, respectively. However, based on the existing literature we can get to know that the higher the rotational speed, the more it affects the transmission error [29]. Analyzing in depth the angular acceleration data at the input in Fig. 18 and Fig. 24, we can find that

to ensure the same joint output angle in the same cycle, the time of the acceleration and deceleration phases of the two variable-speed drive laws increases with the rotational speed, which leads to the reduction of the maximum angular acceleration of the two drive laws by 86.5% and 86.4%, respectively. This is the main reason for the significant reduction in the transmission errors.

Figure 29(c) represents the range of transmission error variation of the RV reducer with the SAD driving law and the EAED driving law respectively at the maximum angular acceleration of $11,000^\circ/\text{s}^2$ and load inertia of 14.35 kg m^2 . According to the data in Fig. 26, the fluctuation range of the transmission error of the RV reducer under the SAD driving law is $(2.008', -1.213')$, and the transmission error is calculated to be $3.221'$; under the EAED driving law, the fluctuation range of the transmission error is $(1.878', -1.955')$, and the corresponding transmission error is $3.833'$. Compared with the SAD driving law, the error value increases by $0.612'$. When the values of maximum angular acceleration are the same, the values of transmission error are similar for both variable speed drive laws, again indicating that the transmission error is most sensitive to angular acceleration.

Figure 29(d) represents the range of transmission error variation of the RV reducer with SAD driving law and EAED driving law respectively at the maximum angular acceleration of $20,000^\circ/\text{s}^2$ and load inertia of 14.35 kg m^2 . According to the data in Fig. 27, the fluctuation range of transmission error of the RV reducer under the SAD driving law is $(2.006', -2.341')$, and the transmission error is calculated to be $4.347'$; under the EAED driving law, the fluctuation range of transmission error is $(2.739', -2.502')$, and the corresponding transmission error is $5.241'$. Compared with the SAD driving law, the error value increases by $0.894'$. When the load inertia is the same and the angular acceleration increases by 81.8%, the transmission error for the SAD drive law increases from $3.221'$ to $4.347'$, for the EAED drive law increases from $3.833'$ to $5.241'$. Hence, the load inertia is the same and the value of transmission error increases with angular acceleration. This phenomenon further confirms that angular acceleration is an important factor affecting transmission error.

Therefore, load inertia and angular acceleration are the critical factors affecting the transmission error of the RV reducer. By comparing the transmission error data of RV reducers with two different driving laws, it can be obtained that, considering the influence of load inertia, the SAD driving law is better than the EAED driving law in terms of control accuracy. Moreover, the larger the angular acceleration is, the larger the difference between the transmission error values of the two variable speed drive laws is.

7 Conclusions

This study analyzes the dynamic transmission error characteristics of RV reducers for industrial robot joints under variable operating conditions. The mathematical models of two types of joint driving laws, commonly used SAD and EAED for industrial robots, are given, and the contact dynamics model of the RV reducer for industrial robot joints considering the geometric errors of critical transmission parts is established. Dynamic transmission error tests of the RV reducer under equal speed working conditions were carried out to verify the correctness of the theoretical model, and then to explore the dynamic transmission error characteristics of the RV reducer by simulating the dynamic behavior of industrial robot joints under variable working conditions. The main findings obtained through the research are as follows:

(1) Without considering the effect of load inertia, the magnitude of the dynamic transmission error of the RV reducer of the industrial robot under the conditions of SAD and EAED

drive is larger than that under the conditions of equal speed drive, with the magnitude of the transmission error under the SAD drive increasing by 6.3%, and that under the EAED drive increasing by 5.1%. The transmission errors under the three driving laws are all within $1'$, and the differences are not particularly obvious.

(2) Load inertia is a critical factor affecting transmission errors in the RV reducer. Considering the effect of load inertia, the magnitude of the dynamic transmission error of the RV reducer of the industrial robot under the conditions of SAD and EAED drive is significantly larger than that under the conditions of equal speed drive, with the magnitude of the transmission error under SAD drive increasing by 494.3%, and the magnitude of transmission error under EAED drive increasing by 599.9%. When the rotational speed is the same and the load inertia increases by 28.2%, the transmission error for the SAD drive law increases from $8.208'$ to $10.732'$, for the EAED drive law increases from $9.666'$ to $11.613'$. Hence the load inertia has a greater impact on the transmission error.

(3) The angular acceleration is also a critical factor affecting the transmission error of the RV reducer. When the load inertia is the same and the rotational speed is increased by 50%, the transmission errors of the two drive laws are reduced by 61.6% and 67.2%, respectively. However, in-depth analysis reveals that to ensure the same joint output angle in the same cycle, the time of the acceleration and deceleration phases of the two variable-speed drive laws increases with the increase of the rotational speed, which leads to the reduction of the maximum angular acceleration of the two drive laws by 86.5% and 86.4%, respectively, so the angular acceleration has a greater impact on the transmission error relative to the rotational speed. When the load inertia is the same and the angular acceleration increases by 81.8%, the transmission error for the SAD drive law increases from $3.221'$ to $4.347'$, for the EAED drive law increases from $3.833'$ to $5.241'$. Moreover, the larger the angular acceleration, the more obvious the difference in transmission error.

(4) Different variable speed drive laws have different effects on the transmission error of the RV reducer. Without considering the effect of load inertia, the transmission error under the SAD drive increases by 1.2% compared with that under the EAED drive. Considering the effect of load inertia, the transmission error under the SAD driving law is always smaller than the transmission error under the EAED driving law, and the larger the angular acceleration is, the larger the difference between the transmission error values under the two types of variable-speed drive laws is. Therefore, without considering the effect of load inertia, the difference in transmission error between the SAD driving law and the EAED driving law is very small. However, since there is no angular acceleration step phenomenon in the SAD driving law, the SAD driving law is better than the EAED driving law in reducing vibration impacts. Considering the effect of load inertia, the SAD law is always better than the EAED driving law in terms of control accuracy.

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Author contributions Xu. provides theoretical guidance and manuscript revision and Wu. is responsible for simulation analysis, data processing and manuscript writing

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Data Availability No datasets were generated or analysed during the current study.

Declarations

Competing interests The authors declare no competing interests.

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